

Pricing the Priceless: The Financing Cost of Biodiversity Conservation*

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Abstract

Safeguarding biodiversity entails substantial economic costs. We study how markets price fiscal risks arising from local government’s conservation effort by examining the “Green Shield Action,” a regulatory initiative launched in China in 2017 to enforce biodiversity protections in National Nature Reserves. We document that, while improving biodiversity, the initiative raises bond yields for municipalities hosting these reserves. Two channels could explain this effect: regulatory costs from compliance burdens, or punishment from investors with non-pecuniary biodiversity preferences responding to revealed mismanagement. Empirical evidence suggests that regulatory costs – not investor preferences – drive the increased cost of public capital.

Keywords: Alternative Data, Biodiversity Transition, Bonds, China, Sustainability

JEL Classification: Q57, Q58, H74, G12

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1 INTRODUCTION

The ongoing rate of biodiversity loss underscores the intricate nexus between nature and the global economy (Flammer *et al.*, 2025). Since 1900, 477 wild vertebrate species have become extinct, and 30 to 50% of all species are estimated to face the risk of extinction by the middle of this century (Ceballos *et al.*, 2015; IPBES, 2019). Prior studies have evaluated the direct economic costs of biodiversity conservation efforts (Deutz *et al.*, 2020) and the asset pricing implications of biodiversity risk (e.g., Garel *et al.*, 2024; Giglio *et al.*, 2023). However, the financing cost of biodiversity conservation and its implications for financial markets remain less explored (Karolyi & Tobin-de la Puente, 2023).

We contribute to this literature by presenting new evidence from a major policy for biodiversity conservation in China, the Green Shield Action (GSA). GSA is a national regulatory initiative launched by the central government in 2017 to reinforce the safeguard of National Nature Reserves (NNRs). NNRs are legally designated areas for biodiversity conservation managed by local governments. In recent decades, many of the local administrations in charge of such reserves failed to fulfill their responsibilities, allowing economic activities such as mining, tourism, and hydropower generation to occur within their NNRs. In July 2017, the central government launched GSA to identify such violations. Under GSA, the central government began conducting rigorous investigations and exerting pressure on local authorities to act in response to reported violations.

We focus on the impact of GSA on the pricing of municipal corporate bonds (MCBs), the de facto primary form of municipal bond in China. Because the funding responsibility of the conservation effort in NNRs largely falls on local governments, the implementation of the GSA could have led to substantial increases in financial pressure for municipalities containing NNRs.

Our empirical analysis relies on three main datasets. First, we manually construct a new dataset that contains the geographical information of all NNRs in China. This allows us to identify municipalities whose boundaries encompass NNRs. Second, we utilize data on the issuance and trade information on all municipal corporate bonds from January 2013 to June 2022. To assess the impact of GSA on local conservation efforts, we use satellite remote sensing data, as well as information sourced from government procurement documents, newspaper articles, and bird observation records.

We implement a standard difference-in-differences (DID) empirical strategy. We compare changes in the yield spreads of MCBs of municipalities containing at least one NNR (NNR municipalities) with those of municipalities without NNRs (non-NNR municipalities) around the launch of GSA. We document that, compared to non-NNR municipalities, NNR municipalities experience a 22.5-basis-point larger increase in their yield

spreads following the introduction of GSA at the end of 2017. This corresponds to 11.5% of the in-sample mean (195.8 basis points).¹ The results are robust to using more flexible measures of treatment intensity (i.e., the area of NNRs), alternative bond spread measures, and data frequencies.

A key challenge with our identification strategy is whether the documented differences in bond yields around the introduction of GSA are driven by differences in expected local public financing costs related to biodiversity conservation or other confounding factors. A comparison of municipality characteristics shows that municipalities with and without NNRs are similar across a large set of observables in the pre-GSA period. Including controls for bond- and city-level characteristics in our estimating equation leaves the size of the main coefficients stable. Event studies show that municipalities with and without NNRs display similar trends in yield spreads in the period before the introduction of GSA, lending support to our identification assumptions.

To interpret our findings, we consider potential channels through which GSA could affect bond pricing. The first is a regulatory cost channel: GSA created credible enforcement that imposed transition costs on local governments, and investors anticipating these fiscal burdens demand higher yields. Consistent with this channel, we document that NNR municipalities experience a significantly larger increase in their fiscal deficit compared to non-NNR municipalities following the implementation of GSA. This effect is primarily driven by increased government expenditures, with a smaller contribution from decreased revenues. To more precisely link GSA with changes in local public expenditure, we collect and analyze the text of local public procurement announcements and identify those linked to nature reserves and biodiversity conservation. We document that, after GSA implementation, NNR municipalities experienced a larger increase in both the number and monetary value of such procurement contracts.

An alternative mechanism is an investor preferences channel: if investors hold non-pecuniary preferences for biodiversity, GSA may have revealed information about which municipalities had poorly managed their reserves, triggering higher required returns as punishment for environmental mismanagement.

These channels yield distinct predictions. Under the regulatory cost channel, effects should scale with the magnitude of ex-ante violations but should not depend on prior information availability. Under the investor preferences channel, effects should be larger where GSA revealed new information—that is, where prior media coverage was low and information surprise was greatest. Utilizing remote sensing data on developed land and

¹ This effect is comparable in scale to the difference in municipal bond yields between states allowing municipal bankruptcy and those that do not (Gao *et al.*, 2019), as well as the rise in credit spreads linked to a one-standard-deviation increase in properties exposed to six feet of sea-level rise (Goldsmith-Pinkham *et al.*, 2023), both roughly 10% of the average sample spread.

nighttime luminosity, we find that a higher presence of economic activities within NNRs before GSA is associated with a more pronounced pricing effect. This is consistent with regulatory costs scaling with expected compliance burdens. Yet effects do not vary with prior newspaper coverage of reserves. This pattern indicates that GSA’s impact operates predominantly through anticipated fiscal costs—either because investors already knew which NNRs were poorly managed, or because investors with non-pecuniary biodiversity preferences are not marginal in this market.

To study the real biodiversity benefits of GSA, we collect data on bird species recorded by bird-watching enthusiasts and research observation stations, as well as species richness measures from the IUCN Red List. We find evidence consistent with a positive effect of GSA on biodiversity. Given these positive ecological outcomes, if investors value long-term sustainable value of GSA, which is not necessarily pecuniary, then one would expect a lower increase in MCB spreads in municipalities where GSA led to a larger improvement in biodiversity. However, we do not find evidence of such heterogeneous effects. This result may stem from the potentially limited number and less influential trading activity of investors caring for biodiversity. The market outcomes we document suggest a misalignment between biodiversity benefits and investors’ valuations, which could hinder broader conservation efforts.

Finally, to quantify the aggregate financial cost of the biodiversity transition, we perform a back-of-the-envelope calculation comparing counterfactual annual interest payments inferred from our estimates with actual payments. Our calculation is based on the outstanding debts of Local Government Financing Vehicles (LGFVs) with MCBs from 2013 to 2021.² We find that GSA led to aggregate extra interest costs of around 40 billion U.S. dollars from 2018 to 2021. As a reference, recent estimates suggest that China would need to spend 131.54 billion dollars per year to protect biodiversity according to its proportion of GDP in the world (Deutz *et al.*, 2020). The extra financing costs brought by GSA account for 4.21%, 8.75%, and 16.24% of these estimated direct costs in 2019, 2020, and 2021, respectively.

Our paper contributes to the literature on biodiversity and nature conservation.³ The literature has documented benefits of biodiversity such as boosting agricultural productivity (Frank *et al.*, 2024), reducing human health risks (Frank & Sudarshan, 2024; Frank, 2024; Keesing & Ostfeld, 2021), enhancing resilience to climate change and natural disasters (Isbell *et al.*, 2015; Rizzi, 2024; Taylor & Druckenmiller, 2022; Liu *et al.*, 2025), and

² In practice, LGFVs are direct issuers of MCBs (see details in Section 2.3).

³ Seminal contributions by Weitzman (1992, 1993, 1998) measure the “value of diversity”; Heal (2000) provides an overview of biodiversity’s services, and Brock & Xepapadeas (2003) integrate ecological and economic frameworks for valuing biodiversity.

improving amenity (Cornaggia *et al.*, 2025; Han *et al.*, 2025). Meanwhile, there is limited evidence on the economic costs associated with biodiversity conservation. Recent work has documented that conservation efforts may lead to unemployment (Ferris & Frank, 2021), business contraction (Akbari *et al.*, 2025) and depreciation of resource extraction rights (Bošković & Nøstbakken, 2017; Bahrami *et al.*, 2024; Frank *et al.*, 2025). Our paper provides empirical evidence on the effects of biodiversity conservation on public financing costs in the context of a large government-led effort, suggesting that the financial market prioritizes conservation costs and credit risk over biodiversity’s sustainability benefits.

Our paper is also related to the emerging literature on biodiversity and financial markets. Recent work includes Giglio *et al.* (2023, 2024), Garel *et al.* (2024, 2025), Coqueret *et al.* (2025), Xin *et al.* (2023), Xiong (2023), Colesanti Senni *et al.* (2025) and Guidolin & Pedio (2025) characterizing biodiversity risks and their risk premia in equity or commodity markets. We extend the discussion by exploring how biodiversity conservation costs are priced in bond markets in a large emerging economy of significant importance for global biodiversity conservation.⁴ Unlike other economic effects that often stay local, external validity is not a concern here given the global nature of sustainability issues such as climate change and loss of biodiversity (e.g., Bolton & Kacperczyk, 2023); if anything, studies beyond settings of developed countries where empirical finance research has focused on prove crucial. Moreover, because most governments do not use equity financing that gives investors an upside in the long run (Starks *et al.*, 2025), our findings point to a particular challenge of funding biodiversity conservation via debt markets.⁵

More broadly, our study is related to the growing body of research on sustainability and climate finance (e.g., Hong *et al.*, 2020; Krueger *et al.*, 2020; Giglio *et al.*, 2021; Edmans & Kacperczyk, 2022; Seltzer *et al.*, 2022; Acharya *et al.*, 2023; Hong *et al.*, 2023; Hsu *et al.*, 2023; Sautner *et al.*, 2023; Li *et al.*, 2024; Ramadorai & Zeni, 2024; Addoum *et al.*, 2024; Duchin *et al.*, 2025). Relative to research on climate, biodiversity research lacks reliable and widely accepted evaluation measures, making it highly challenging to define property rights related to conservation obligations for the private sector (Karolyi & Tobin-de la Puente, 2023; Gjerde *et al.*, 2025). These characteristics make it difficult to design market-based mechanisms for the “monetization” of biodiversity value similar

⁴ For instance, Egli *et al.* (2018) find that optimizing integrated land use planning for sustainable intensification in ten countries—China, India, the Philippines, Brazil, Australia, Mexico, Indonesia, the Democratic Republic of the Congo, Ecuador, and Vietnam (with a majority being emerging economies)—could prevent an estimated one-third of biodiversity loss by 2040.

⁵ Economists have been increasingly focusing on the impact of biodiversity risk on credit risk, specifically physical and regulatory risks faced by private entities (Hoepner *et al.*, 2023; Soylemezgil & Uzmanoglu, 2024; Arlt *et al.*, 2024; Nguyen *et al.*, 2025; Deng & Lin, 2025). Our study directly examines the impact of government-led transition investments on the cost of public capital. This perspective complements, yet differs from, the private sector’s risk exposure, especially amid the ongoing evolution of blended finance (Karolyi & Tobin-de la Puente, 2023; Flammer, 2021; Cao *et al.*, 2025).

to carbon trading, thus necessitating direct public capital intervention to address market failures related to common goods (Kedward *et al.*, 2022). Therefore, in biodiversity transition, the role of government is particularly important and the impacts of biodiversity risks on the public sector extend beyond revenue risks and asset value depreciation (volatility) (Painter, 2020; Auh *et al.*, 2022; Goldsmith-Pinkham *et al.*, 2023; Cornaggia & Iliev, 2023; Jerch *et al.*, 2023; Jha *et al.*, 2025)—exactly what our work underscores.

In addition, our paper also sheds light on investors’ considerations of sustainability value (Starks, 2023). Previous studies suggest positive financial outcomes of sustainable practices (e.g., Pástor *et al.*, 2022; Ardia *et al.*, 2023; Flammer, 2021; Baker *et al.*, 2022; Feldhütter *et al.*, 2024; Krüger, 2015) and investigates investors’ underlying motivations (e.g., Baker *et al.*, 2024; Barber *et al.*, 2021; Giglio *et al.*, 2025; Heeb *et al.*, 2023). We find that responsible investments focusing on biodiversity value remain insufficient to significantly influence equilibrium prices during GSA. This provides evidence and motivation for research on distinct features of investor preferences and incentive-compatible financing tools in biodiversity domains.

2 INSTITUTIONAL BACKGROUND

2.1 NATURE RESERVES IN CHINA

With its diverse climate and large number of recorded known species — 138,293 (by 2022), 6.4% of the total number recorded worldwide — China is among the most biologically diverse countries in the world.⁶ For example, China is the third country by number of vascular plant species, after only Brazil and Colombia. This wealth of biodiversity underscores its vital importance for the global ecosystem.

To protect its natural capital, China has designed an extensive network of nature reserves — land, inland waters, or seas containing representative ecosystems, habitats for rare and endangered wild species, or natural relics with special significance.⁷ Among them, National Nature Reserves (i.e., NNRs) have the most extensive regulatory oversight and strictest legal provisions compared with other provincial, municipal, and county nature reserves. Correspondingly, the regulations stipulate that only reserves “holding general significance domestically and internationally, exerting remarkable international scientific influence, or possessing extraordinary research value” can be designated as NNRs.

⁶ See, e.g., <http://www.sp2000.org.cn/CoLChina> and <https://www.iucnredlist.org/resources/summary-statistics>.

⁷ Our definition of nature reserves follows the law and regulation of the People’s Republic of China that govern the establishment and management of nature reserves within the territory and other sea areas under the jurisdiction of the country. In particular, the regulation stipulates that nature reserves are for biodiversity preservation.

The designation of NNRs is solely based on the long-term endowment of natural resources, without considering contemporaneous local economic and fiscal conditions of the regions where the reserves are located.⁸ According to regulations, the local government should first submit a formal application to the administrative authority of NNRs under the State Council. The central review committee for NNRs is responsible for the evaluation process. After receiving a report on the biodiversity situation of the candidate reserve, the committee members vote. For an NNR to be eligible for State Council approval, it must receive a two-thirds majority vote. Evaluation criteria include fragility, diversity, endangered status, and scientific value of the natural resources of the reserve. The members of the review committee are predominantly experts in the field of natural sciences, with no representation from the economic or financial domains.

Following the establishment of an NNR, the local governments are responsible for managing, supervising, and conserving NNRs within their jurisdictions. Each NNR is typically governed by a dedicated committee that manages its operations. This committee functions as an administrative entity under the local government’s authority, and it is financed by local government funds.⁹ NNRs also receive funding from the central government. However, this central funding is usually limited. For example, Jianbo Sun, a deputy of the 13th National People’s Congress, presented a proposal highlighting that the central government’s annual funding for all NNRs totals only 600 million CNY (about 80 million USD). This breaks down to an average of just 1.26 million CNY (about 170 thousand USD) for each reserve. Given limited central funding, local governments bear most of the financial responsibility for maintaining NNRs.¹⁰

As Figure 1 shows, the three decades between 1980 and 2010 have seen a large increase in the establishment of new nature reserves, whose number has stabilized during the last decade.¹¹ In the early decades, the effectiveness of nature reserves in protecting biodiversity fell short of expectations. Local officials prioritized economic development over environmental issues (Guan *et al.*, 2010; Liu, 2010), resulting in the deterioration of numerous reserves. A prime example is the Qilian Mountain NNR in Gansu Province,

⁸ Evaluation criteria for NNRs are available at https://www.mee.gov.cn/stbh/zrbhq/gjjzrbhqps/201605/t20160522_342427.shtml.

⁹ See Appendix B.3 for more details regarding the institutional structure arrangement of NNRs.

¹⁰ See Appendix B.4 for more details on funding sources for nature reserves.

¹¹ The history of reserves dates back to the 1950s, with the establishment of the first nature reserve, the Dinghushan Nature Reserve. Following the economic reform and liberalization of 1978, the number of nature reserves underwent substantial growth. Toward the end of the 20th century, China also experienced several environmental crises, including the drying out of the Yellow River in 1997, the flooding of the Yangtze River in 1998, and the sandstorm that swept Beijing in 2000. These disasters necessitated a reconsideration of the connection between economic growth and ecology, culminating in a further significant expansion of nature reserves during 1997-2010. After 2010, the rate of growth of nature reserved slowed. Currently, there are over 2,700 nature reserves (around one-sixth of which are NNRs), covering more than 140 million hectares.

where exploration and mining activities caused substantial damage to local vegetation, soil erosion, and surface subsidence. For instance, 14 mining and exploration projects within this area were found to be unauthorized, with 3 of these instances located within the core area of the NNR, where in principle no individual is allowed to enter and no economic activity can be carried out. In addition, over 30 mining projects were concealed, and over 40 hydropower plants were constructed illegally.¹² Following the “Qilian Mountain Incident,” the Ministry of Environmental Protection of China announced plans to inspect environmental issues in all national nature reserves from July to December 2017. The initiative was named the Green Shield Action (GSA).

2.2 GREEN SHIELD ACTION

Launched in July 2017, GSA is a major regulatory effort by the central government to enforce biodiversity conservation in all NNRs. Its initial round identified over 20,800 potential issues related to nature reserves, resulting in the closure and ban of over 2,460 enterprises and the demolition of over 5.9 million square meters of constructed facilities. Meanwhile, more than 1,100 local government officials were held accountable, and several local regulations inconsistent with higher-level laws were modified. Overall, GSA marks a clear change in China’s effort to regulate and supervise NNRs, and reflects policymakers’ commitment to improving biodiversity conservation.

In practice, GSA is a special supervision campaign implemented via investigation teams in charge of on-site inspections of national nature reserves. Investigation teams use various technologies – including high-resolution remote sensing – to inspect NNRs, identify illegal activities, and report to the central government. Following these reports, the central government urges local administrations to investigate the uncovered violations and rectify them. Every local government unit in charge of an NNR must also create and execute a detailed work plan that aligns with the guidelines set by GSA. This plan includes regular inspections. Through these approaches, the central government placed significant political incentives on local officials to ensure that illicit activities are curbed.¹³

GSA implementation encountered local resistance. First, several NNRs are severely understaffed, making it difficult to comply with requests from the central government. Moreover, prior to the designation of an area as an NNR, there was already a large number of residents living within these regions. Relocating residents and demolishing existing structures is extremely challenging. Finally, local governments also need to

¹²The central government conducted a special investigation into the Qilian Mountain National Nature Reserve, verified the ecological damage, held local officials and state-owned enterprises accountable, and made a public statement (see http://www.gov.cn/xinwen/2017-07/20/content_5212107.htm).

¹³See [Appendix B.5](#) for further details regarding the political incentives faced by local officials.

undertake ecological restoration projects within the reserves and allocate substantial funds to enhance the supervision of these areas and improve their operations.¹⁴

An illustrative case is the relocation of the Jinyun Mountain NNR. In principle, the reserve was strictly off-limits to all individuals and entities, with only personnel engaged in scientific research and observation activities being legally allowed to enter certain zones. However, the reserve had nearly 9,000 residents living within its boundaries. This imposed a critical burden on local authorities that had to evacuate the area to meet GSA’s requirements. Specifically, local government was estimated to invest at least 800 million CNY (over 100 million USD) in the relocation efforts for the Jinyun Mountain NNR during 2018 - 2020. Furthermore, GSA stipulates that the local government must demolish all illegal buildings within the reserve, including housing, rural tourism facilities, hotels, and horse farms. Such requirements further compound the economic costs.¹⁵

After the first phase of GSA in 2017, the central government continued its efforts in subsequent years by verifying the implementation of rectification requirements, scrutinizing new violations of laws in nature reserves, and supervising the fulfillment of management responsibilities of relevant departments. These efforts indeed promoted the restoration of the local ecology (e.g., see the changes in the land cover of the Qilian Mountain NNR shown in [Figure A1](#)). By perpetuating these actions after 2017, the central government solidifies the expectations regarding its commitment to protect biodiversity and nature reserves, thus shaping the financial market’s perception of the transition costs associated with biodiversity conservation.

2.3 MUNICIPAL CORPORATE BONDS

In the 1990s, the tax-sharing reform in China substantially reduced the proportion of tax revenue allocated to local governments. Meanwhile, the 1994 Budget Law prohibited local governments from directly engaging in any form of debt financing. For municipalities, one way to balance the growing demand for public investments and insufficient fiscal resources is to establish local government financing vehicles (LGFVs), special-purpose state-owned enterprises. LGFVs primarily undertake the supply of local common goods (e.g., public infrastructure). To finance these public projects with long cycles and low direct earnings, LGFVs can borrow from financial institutions and issue bonds, backed by local-authority-appropriated land, subsidies, and other implicit guarantees.

Due to regulatory restrictions, LGFVs grew slowly in the early stage until the central government launched a large stimulus package in 2008 in response to the global financial

¹⁴See [Appendix B.6](#) for further details regarding the challenges encountered during GSA.

¹⁵These economic costs at least include: (1) Relocation of indigenous residents. (2) Demolition of illegal structures. (3) Ecological restoration. See [Appendix B.7](#) for further details regarding the initiatives undertaken by the Chongqing government.

crisis. The fiscal part of the package is commonly known as the 4-trillion-CNY plan, which mainly consists of public infrastructure and social welfare projects. In terms of financing, more than two-thirds of planned investments (2.82 trillion CNY) were expected to be funded by local governments. To facilitate these programs, the central government introduced a series of credit expansion (Cong *et al.*, 2019) and financial deregulation policies (Bai *et al.*, 2016), encouraging local governments to raise funds through LGFVs.¹⁶ Subsequently, the liability of LGFVs experienced a significant surge, which raised concerns about the default risks of municipalities.¹⁷ Then Beijing reverted its aggressive credit policy back to normal in 2010, making LGFVs resort more to bond financing when facing rollover pressure from bank debt coming due around 2012 (Chen *et al.*, 2020).

The bonds issued by LGFVs are generally referred to as municipal corporate bonds (MCBs, see, e.g., Chen *et al.*, 2020; Gao *et al.*, 2021),¹⁸ where “corporate” reflects the fact that LGFVs have the same legal status as other regular corporations nominally, and “municipal” indicates the exclusive implicit guarantee from local government. In 2015, the new Budget Law became effective, allowing provincial governments to issue municipal bonds directly. However, MCBs are still the dominant form of local public financing. The overwhelming majority of city governments are not authorized to issue municipal bonds on their own.¹⁹ Most MCB issuers are LGFVs owned by local governments below the province level, providing us with variation in bond prices at a fine geographical level.²⁰

Importantly, the market holds faith in municipalities’ implicit guarantee for LGFVs and thereby prices credit risk of local governments when investing in MCBs (Liu *et al.*, 2017). In general, MCBs’ credit rating reports typically consider local governments’ fiscal conditions first, which is rarely seen among other issuers. For example, in Moody’s Local

¹⁶Financial deregulation policies include: (1) Guidelines on Further Strengthening the Adjustment of Credit Structure to Promote Steady and Rapid Development of Economy, released by the China Banking Regulatory Commission (CBRC) in March 2009, (see http://www.gov.cn/gongbao/content/2009/content_1336375.htm). (2) the Notice on Accelerating the Implementation of Local Supporting Funds for Central Government Investment Projects to Expand Domestic Demand, released by the Ministry of Finance of the People’s Republic of China (PRC) in October 2009 (see http://www.gov.cn/zwzk/2009-10/13/content_1437713.htm).

¹⁷Fan *et al.* (2022) show that in 2009, the total value of LGFV bonds increased by 217%, and bank loans to LGFVs surged by 93%. In addition, the debt raised by LGFVs represented more than 70% of the total debt of local governments, and it surged from 1.7 trillion CNY in 2007 to 6.6 trillion CNY in 2010 and doubled again in 2014. Gao *et al.* (2021) show that some LGFVs led to actual defaults on loans, suggesting that municipalities indeed have considerable credit risk that could be priced in MCB yields, consistent with Ang *et al.* (2023).

¹⁸MCB is “cheng-tou-zhai” in Chinese, meaning “city investment bond,” synonymous with other translations, such as “local government bond” (Huang *et al.*, 2020), “Chengtou bond” (Ang *et al.*, 2023), “urban construction and investment bond” (Liu *et al.*, 2017), and “LGFV bond” (Fan *et al.*, 2022).

¹⁹According to regulatory requirements, only Dalian, Qingdao, Ningbo, Xiamen, and Shenzhen have the right to issue municipal bonds independently among the hundreds of prefecture-level cities.

²⁰In our sample, bonds issued by LGFVs owned by provincial governments only account for approximately 10% of the total MCBs from 2013 to 2022.

Government Financing Vehicles in China Methodology, “government support” is set as the primary evaluation factor of MCB rating.²¹ The rationale behind this is that since cities do not have discretion on issuing municipal bonds directly, their LGFVs still rely on financial support from local governments. Governments at different levels have repeatedly defused the debt repayment crisis of LGFVs, and there have been no real defaults on MCBs so far.²² Given this institutional setting, the price of MCB could reflect the market views on local public financing costs during our sample period.

3 DATA DESCRIPTION

3.1 NATIONAL NATURE RESERVES

We collect the list of nature reserves and zoning images of NNRs from the Ministry of Ecology and Environment of the People’s Republic of China (PRC). By aligning zoning images of NNRs with the standard administrative map of China, we construct a dataset that includes information on NNR locations, borders, inception dates, and tier classifications of NNRs. This dataset enables us to calculate some fundamental attributes of NNRs (e.g., land area) at both the reserve and administration levels. Furthermore, we can construct some novel metrics (e.g., human economic activities within NNRs) after merging the data with other geographical datasets. [Figure 2](#) presents the geographical distribution of NNRs as of 2016 (i.e., the year prior to the GSA shock).

3.2 DATA ON MUNICIPAL CORPORATE BONDS

We obtain municipal corporate bond data from Wind Information Co. (WIND), a leading financial data vendor in China and the most widely used MCB database for our sample period. WIND provides data on: (1) The issuing yield and attributes of each MCB, including issuing price, issuing yield, issuance date, maturity date, issuing amount, bond type, interest type, credit rating at issuance, state of guarantee, and option clause. (2) Daily transaction information of each MCB, including trading price, trading yield, trading date, trading volume, trading site, residual maturity, and credit rating.²³ (3) Information on each MCB issuer, including location, affiliation, and bond issuance. (4)

²¹See <https://ratings.moodys.com/rmc-documents/386644> (This methodology was first published on July 29, 2020, and updated on April 12, 2022).

²²In contrast, some credit bonds issued by private-owned and non-LGFV state-owned enterprises have defaulted ([Geng & Pan, 2024](#)).

²³Consistent with what [Geng & Pan \(2024\)](#) and [Fang et al. \(2025\)](#) described, WIND reports historical transaction information in the daily frequency, with price-related metrics weighted by the volume. In addition, “trading day” refers to the days on which real transactions took place on the corresponding bond, excluding the days on which investors could trade but did not.

Daily yield curve of Treasury bond and the Chinese Development Bank (CDB) bond.²⁴

Following [Geng & Pan \(2024\)](#), we study MCB yields around GSA at the quarterly frequency using the last daily transaction price of the quarter for each MCB. To proxy for the risk premiums that investors demand for investing in bonds, we calculate MCB spreads as the difference between the MCB yield and CDB yield on the same day and of the same maturity ([Chen *et al.*, 2023](#); [Geng & Pan, 2024](#)).²⁵ Treasury yield benchmark is employed in robustness checks.

3.3 SAMPLE AND SUMMARY STATISTICS

Our sample covers the period from January 1, 2013, to June 30, 2022.²⁶ [Table 1](#) reports the descriptive statistics for the main variables used in our analysis. Besides MCB spread and relevant indicators of GSA described above, there are other bond- and city-level variables. Bond characteristics include the logarithm of the issuing amount, residual maturity, bond rating, issuer rating, and whether the bond is option embedded, guaranteed, and traded on the exchange market. City-level characteristics include baseline variables capturing the economic and social conditions of a city collected from the National Bureau of Statistics and local statistical yearbooks in China in 2013. [Table A1](#) provides the details of all the variable definitions. We winsorize MCB spreads and city-level continuous variables at 1% on each tail of the distribution.

[Table 1](#) shows that about 51% of MCB observations are from NNR municipalities, with about 63% of the observations occurring after the launch of GSA. MCBs in our sample have an average residual maturity of 3.83 years and tend to be traded mostly in the interbank (over-the-counter) market. For credit ratings, we convert letter grades into numerical numbers by assigning 1 to AAA, 2 to AA+, 3 to AA, and so on. The average bond rating and issuer rating are below 3. The relatively safe rating level reflects the “municipal” nature of MCBs.

²⁴We restrict the bond sample to MCBs issued by city- and county-level LGFVs with fixed interest rates and residual maturity above one year. We also exclude MCBs in the form of private placement notes, convertible bonds, exchangeable bonds, and asset-backed securities, due to their non-standard structure, limited market size, or insufficient information disclosure.

²⁵The CDB is China’s largest development-oriented financial institution with a safe degree of creditworthiness ([Gao *et al.*, 2021](#)). Besides, tax policy plays a critical role in municipal bond pricing ([Babina *et al.*, 2021](#)). The same as MCBs, CDB bonds are not tax-exempt.

²⁶As explained by [Chen *et al.* \(2020\)](#), the MCB market was relatively underdeveloped before 2012. Another vital fact is that since the 18th National Congress of the Communist Party of China (CPC, also referred to as CCP) at the end of 2012, China overall has experienced a new stable political cycle in our sample period without the turnover of the general secretary of the CPC.

4 EMPIRICAL STRATEGY AND FINDINGS

4.1 EMPIRICAL STRATEGY

Our baseline empirical strategy is a standard difference-in-differences approach that compares the relative changes in the MCB spreads in NNR municipalities and non-NNR municipalities around the introduction of GSA. The specification takes the following form:

$$Spread_{bict} = \delta_{ic} + \delta_t + \beta NNR_c \times Post_t + \theta X'_c \times Post_t + X'_{bict} \gamma + \varepsilon_{bict}, \quad (1)$$

where b indexes bonds, i indexes issuers, c indexes cities, and t indexes year-quarters. The dependent variable, $Spread_{bict}$, is the spread of bond b issued by issuer i , located in city c , and traded in year-quarter t . NNR_c is a dummy variable that equals one if a city geographically intersects with an NNR and zero otherwise. $Post_t$ is a dummy variable that equals one for the period after the introduction of GSA (i.e., from the third quarter of 2017 to the second quarter of 2022). We also include bond issuer fixed effects and year-quarter fixed effects. X_c denotes city-level control variables. Specifically, we interact the pre-determined city-level variables observed in 2013 with $Post_t$ to control for differential trends across cities with different initial characteristics. We control for time-varying bond-level characteristics (X_{bict}) and cluster standard errors at the city level.²⁷

One obvious concern when estimating Equation (1) is that NNR and non-NNR municipalities might differ along other dimensions, and, in particular, dimensions that could affect the evolution of their bond prices in the period around the launch of GSA. To test this concern, we collect data on city-level baseline characteristics from the National Bureau of Statistics and local statistical yearbooks and compare NNR and non-NNR municipalities across a set of observable characteristics including measures of local economic development (log GDP per capita, GDP share of the tertiary sector, log nightlight intensity, log average housing prices, urbanization rate, share of adults with at least high-school education), size (log population and log GDP) and the fiscal position of local governments. We compare NNR and non-NNR municipalities along these characteristics both in levels (at the beginning of our sample period) and in changes (between 2013 and 2016, in the period before the introduction of GSA). The results of the balance test are reported

²⁷In 2017 and 2018, China introduced 29 new NNRs, which were upgraded from provincial nature reserves. No additional NNRs have been added since 2019. These new NNRs account for only 6% (29/475) of the total number of existing NNRs and encompass 22 cities. In addition, 2 of non-NNR municipalities established national parks in 2021. Although new NNRs have also been under the supervision of GSA since their establishment, we are still concerned about the potential selection bias associated with the establishment of these NNRs. To ensure the validity of our empirical identification, we exclude all 24 cities where the number of nature reserves at national level or above changed after GSA. Thus, the presence of NNRs are completely pre-determined in our sample.

in [Table 2](#). As shown, NNR and non-NNR municipalities are comparable in 21 out of 24 observable characteristics in the period before the introduction of GSA. Statistically significant differences emerge in terms of average annual GDP growth rate in the pre period, with NNR municipalities growing on average 0.5 percentage points slower than non-NNR municipalities. This corresponds to 5% of the average growth, as average local GDP growth was 10.2 p.p. for NNR municipalities and 10.8 p.p. for non-NNR municipalities. We also find small but significant differences in average growth in log nightlight intensity (2.4% higher for NNR municipalities), and population growth (1.2% lower for NNR municipalities). In the empirical analysis we control for municipality characteristics and show that magnitudes of the main effects of GSA on spreads remain stable when including such controls.

4.2 MAIN RESULTS

[Table 3](#) reports the baseline estimates for [Equation \(1\)](#). Column (1) includes issuer and time fixed effects to capture time-invariant city and issuer characteristics, as well as common time shocks. In Columns (2)-(4), we augment the specification with bond- and city-level controls. The point estimates across all specifications are positive and statistically significant at standard levels. The estimates remain stable in magnitude when including controls, consistently with the balance table results. In terms of magnitude, the estimates in Column (4) indicate that NNR municipalities experienced a larger increase in MCB spreads of around 22.5 basis points following GSA compared with non-NNR municipalities. This effect corresponds to a 11.5% increase from the sample mean (195.8 basis points). These findings are consistent with the notion that the costs brought about by GSA are priced in the MCB market.

To investigate pre-existing trends, we estimate a period-by-period specification of the following form:

$$Spread_{bict} = \sum_t \beta_t NNR_c \times q_t + \delta_{ic} + \delta_t + \theta X'_c \times Post_t + X'_{bict} \gamma + \varepsilon_{bict}, \quad (2)$$

where all variables are defined as in [Equation \(1\)](#). The coefficients β_t capture the differences between the treatment and control groups in each quarter between the first quarter of 2013 and the second quarter of 2022.

[Figure 3](#) plots the estimates of [Equation \(2\)](#) and their 95% confidence intervals. The conditional difference between NNR and non-NNR municipalities is small in magnitude and non-statistically significant before the introduction of GSA. This is consistent with the parallel trend assumption. The difference increases in the post-GSA period starting about one year after the introduction of GSA. This delay may appear puzzling as one

would expect bond prices to respond immediately when the policy is announced. There are at least two factors to the lag. First, the central government inspected NNRs between late 2017 and early 2018, and only after that – and violations identified and punished – did local governments start planning serious responses and announced their response plans. It is plausible that, at the time of the GSA announcement, investors lacked information about the direct expenditures required to address them, which were primarily disclosed a year later with the fiscal budget statements in early 2019.

Second, it is plausible that both local officials and investors were initially uncertain about the true commitment of the central government to the reform (Wang *et al.*, 2023). Each year, local officials receive multiple campaign-style requests from Beijing, only a few of which represent true priorities for the central government. As enforcement intensified and annual inspections became the norm, it became evident that compliance with this policy was expected. This shift from campaign-style request to serious enforcement could have led investors to only gradually recognize that the exposed local governments would incur significant biodiversity transition costs. We further discuss the timing of the documented effects in [Appendix B.7](#) using a case study.

[Figure 3](#) shows that the gap in the MCB spreads between NNR municipalities and non-NNR municipalities was not a temporary increase followed by a decline, but rather a sustained rise. This pattern aligns with the features of GSA enforcement described above. Inspections on NNRs started in 2017 and were repeated annually in subsequent years. Under such a regular supervisory arrangement, it was difficult for local officials to undertake only temporary solutions, and the true magnitude of the necessary ecological restoration projects became more and more clear over time. In this sense, the process of implementation of GSA guidelines by local governments was not immediate but rather gradual and progressive. [Appendix B.7](#) presents a detailed case in which a local government adopted a phased investment plan following GSA.

To provide additional evidence that the documented effects are indeed driven by the differential impact of GSA on cities with NNRs, we present a placebo test in which we study the differential impact of GSA on cities with nature reserves other than the nationally recognized ones (e.g. provincial, municipal, or county-level nature reserves). Similar to NNRs, these other natural reserves may also restrict business in the way that hinders land use for urban exploitation. Despite this, GSA did not prioritize the inspection of non-NNR reserves. Due to the lack of political incentives, as well as the much larger number of non-NNR reserves than NNRs (approximately fivefold), it is also less likely

that local officials were proactive in rectifying issues in lower-level reserves.²⁸

We augment equation [Equation \(1\)](#) with dummies for the presence of *ProvNR* (Provincial nature reserve), *CityNR* (City-level nature reserve), and *CountyNR* (County-level nature reserve) within the boundaries of a given county, each interacted with *Post*. [Table 4](#) presents the results. We find that variation in non-NNR reserves has no additional effect on bond spreads once NNRs are considered. Overall, the findings support the interpretation that the documented effects are driven by GSA and not by dynamic confounders related to other regional attributes associated with the presence of natural capital.

5 MECHANISMS

5.1 LOCAL PUBLIC CREDITWORTHINESS

The results presented in the previous section can be rationalized by investors' concerns about the effects of GSA-driven transition costs on local public creditworthiness. In this section we study the effect of GSA on local government fiscal deficit and its components. In our sample, no MCB is issued explicitly for biodiversity conservation. Thus, MCB investors may pay exclusive attention to the overall credit condition of the city government behind the MCB. We calculate local fiscal deficit as the difference between public general budget expenditure and revenue, normalized by the pre-shock level of local fiscal revenue or GDP (i.e., the value in 2013). Greater fiscal deficit typically indicates a poorer capacity for public debt repayment. We construct a city-year panel for the period 2013-2021 and test the effects of GSA on local fiscal deficit and its components. We include year and city fixed effects, as well as baseline city-level pre-shock controls interacted with *Post* variable.

[Table 5](#) reports the results. We find that NNR municipalities experienced a significantly larger increase in their fiscal deficit than non-NNR municipalities following the introduction of GSA. Next, we study the impact on government revenues and expenditures (normalized by city GDP value in 2013), both of which enter the fiscal deficit calculation. As shown in Columns (3)-(4) of [Table 5](#), the deterioration in the local fiscal position of NNR municipalities is mostly driven by an increase in expenditure rather than a decline in tax revenues. Specifically, the point estimate in Column (3) indicates that, following the GSA, NNR municipalities experienced an additional increase in fiscal expenditure, comparable to 1.4% of the corresponding city GDP in 2013. Column (4) shows that NNR municipalities also experienced a decline in local government revenues, whose absolute

²⁸Occasionally, GSA extended to natural reserves beyond those officially classified as NNRs. Inspections of non-NNR reserves were often one-off without consistent enforcement efforts in subsequent years. In addition, official statements from the central government focus on revealing the performance of GSA's rectification efforts in NNRs, and contain no mention of natural reserves not classified as NNRs.

value about one-third of the impact on expenditures. Overall, the evidence presented in [Table 5](#) indicates that the transition costs incurred by local governments following GSA stem primarily from higher public expenditure obligations as opposed to revenue drops.

5.1.1 *Government Procurement Contracts for Biodiversity Conservation*

[Table 5](#) shows that the deterioration of local fiscal conditions due to GSA was mostly driven by higher spending. In this section we dig deeper into the nature of local government spending in an attempt to quantify at least part of the direct costs related to biodiversity conservation.

The main challenge is the lack of available data on the exact allocation of fiscal resources across different projects. In particular, we cannot separately observe specific portions of the local government budget dedicated to biodiversity conservation, especially when it comes to general items such as salaries of civil servants responsible for nature reserves and compensation for resettlement of indigenous communities and businesses. To make progress on this front, we analyze local spending via public procurement announcements ([Beraja et al., 2023](#)). These deals provide insights into at least a portion of the direct public investments in protecting biodiversity.

Data on public procurement announcements are sourced from the China Government Procurement website, maintained by the Ministry of Finance. We examined two types of biodiversity procurement, the first being narrow-scope biodiversity procurement associated with NNRs. We define such biodiversity procurement as those where either the title or the purchasing entity of the deal explicitly mentions the name of an NNR.²⁹ The data allows us to identify narrow-scope biodiversity procurement deals from 2015 to 2021.³⁰

We also create a broader definition of public procurement linked to biodiversity initiatives. To identify such procurement, we adopt a machine learning approach, constructing a biodiversity-related keyword dictionary, without requiring announcements to explicitly reference NNRs. Specifically, we first train a Word2Vec model on the full corpus of government procurement documents. Using “biodiversity” and “nature reserve” as seed terms, we then expand the lexicon by extracting semantically similar terms based on vector-space proximity, ultimately generating a biodiversity-related keyword dictionary.³¹ We then identify all procurement deals containing terms from this dictionary and combine them with the narrow-scope biodiversity procurement. After removing duplicates, this

²⁹For instance, a deal for “Remote Monitoring Facility Project of Heixiazi Island National Nature Reserve” would be considered an NNR procurement as it includes the NNR name “Heixiazi Island” in its content.

³⁰We provide examples of biodiversity procurement in [Appendix B.1](#).

³¹The dictionary can be found in [Appendix B.2](#).

process yields our final set of broad-scope biodiversity procurement. We manually extract the monetary value for all deals in both types.

We geocode all purchasers’ addresses using the Amap API, combine them with announcement dates, and then construct a city-year panel dataset. We use biodiversity procurement’ number and value (normalized by city GDP value in 2013) in each city as the dependent variable to examine the impact of GSA. [Table 6](#) presents the results for two types of procurement. We find that both narrow-scope and broad-scope procurement increased significantly after GSA implementation.

The magnitudes of the coefficients imply that for a municipality with GDP in 2013 at the sample mean (around 310 billion CNY, according to [Table 1](#)), NNR municipalities experienced a 4 (narrow definition, 0.014×310) to 24 (broad definition, 0.077×310) more biodiversity procurement deals relative to non-NNR municipalities. In terms of procurement value, the estimates suggest that the increase in procurement spending on biodiversity conservation in NNR municipalities relative to non-NNR municipalities was 0.004% (narrow-definition) to 0.023% (broad-definition) of local GDP in 2013.

In evaluating the magnitude of these effects, it is important to keep in mind that these procurement deals represent only a fraction of the total biodiversity expenditures. The primary aim of our analysis is to identify changes in government spending patterns, specifically increased investments in targeted biodiversity conservation subcategories. This is because not all government expenditures require information disclosure. Only specific products or services (e.g., certain types of machinery, software, technical services, or integrated projects) or transactions exceeding a certain threshold (e.g., 0.5 or 1 million CNY in some regions) are subject to standard bidding procedures and disclosure requirements on procurement websites.³² Activities such as tree-planting compensation, hiring, relocation subsidies, and scientific surveys are exempt from government procurement requirements.

For all these reasons, the share of total conservation expenses captured by procurement documents is likely to be limited. One example illustrates this point. In [Appendix B.7](#), we describe the case of Jinyun Mountain, an NNR in Chongqing. Between 2018 and 2020, following the GSA, the local government invested a total of 2.895 billion CNY in the Jinyun Mountain NNR. The primary allocations were for three purposes: demolishing illegal constructions, relocating indigenous residents, and restoring the local ecological conditions. A substantial portion of these funds was not disbursed through public procurement processes. For instance, local government spent at least 800 million CNY on resident relocation, an expenditure category excluded from government procurement activities. In fact, specific procurement deals related to the Jinyun Mountain NNR during

³²Moreover, when addressing urgent tasks, the complex and time-consuming nature of procurement process incentivizes local authorities to circumvent it, for example, by splitting a deal to multiple ones.

this period amounted to 0.039 billion CNY, representing 1.3% of the total investment of 2.895 billion CNY. This observed procurement expenditure proportion aligns closely with our estimates: as indicated in [Table 5](#), increased biodiversity-related procurement spending constitutes 0.3% (narrow definition, 0.004/1.4) to 1.6% (broad definition, 0.023/1.4) of the reported increase in fiscal expenditure shown in Column (3) of [Table 5](#).

5.2 DRIVERS OF INVESTORS' RESPONSE: REGULATORY COSTS VS PREFERENCES

Two distinct channels could explain the increase in financing costs for NNR municipalities. The first is a *regulatory cost channel*: GSA created credible enforcement that imposed transition costs on local governments – shutting down illegal activities, relocating residents, and funding ecological restoration. Investors anticipating these fiscal burdens would demand higher yields, with effects scaling in proportion to expected compliance costs. The second is an *investor preferences channel*: if investors hold non-pecuniary preferences for biodiversity, GSA may have revealed information about which municipalities had poorly managed their reserves, triggering divestment or higher required returns as punishment for environmental mismanagement.

These channels yield distinct predictions. If investors are aware of management quality ex-ante, then those with non-pecuniary preferences for biodiversity would have already adjusted their positions before GSA is implemented. In this case, the effect of GSA should be purely reflect regulatory costs. On the other hand, if investors were previously unaware of management quality, GSA could trigger both a preferences effect and anticipation of regulatory costs effect, as described in the table below:

	Well Managed NNRs	Poorly Managed NNRs
Known mgm quality	No Effect	Regulatory Cost Effect
Unknown mgm quality	Preferences Effect	Regulatory Cost + Preferences Effect

We test these predictions using heterogeneity in ex-ante economic activity within NNRs as a proxy of management quality, and prior newspaper coverage of NNRs as a proxy for investors' knowledge of such quality.

We use two measures for pre-existing economic activity in an NNRs: the level of developed lands and nightlight luminosity within NNRs in 2016. To identify developed land, we use remote sensing data from the MODIS-IGBP program, which classifies land cover into various vegetation types, wetlands, water bodies, barren lands, urban built-up lands, croplands, and others. We use urban built-up lands and croplands to capture developed lands within NNRs, and raster data on nighttime luminosity from [Zhang et al. \(2024\)](#).

[Table 7](#) reports the results. We augment [Equation \(1\)](#) with a triple interaction

of $NNR \times Post$ and a dummy capturing the top quartile of economic activity within NNRs, plus the additional required terms of the triple-difference model (omitted from the table).³³ We find that the estimated coefficient of the triple interaction term is positive and statistically significant in both specifications. The magnitude of the triple interaction indicates that municipalities with larger pre-existing developed lands within NNRs experienced a two to three times larger impact from the GSA on bond yields.

The findings in Table 7 align with the notion that cities with more illegal activities within NNRs before GSA were more likely to face increased public financing costs. However, these effects could be driven by a combination of preferences and regulatory effects. In particular, some investors' decisions may be motivated by their (non-pecuniary) preferences for sustainability (Starks, 2023). If investors are uninformed about poor mismanagement of NNRs before the GSA, the preferences and regulatory cost effects will both be at work. To disentangle the two, we examine whether GSA impact varies with prior information availability.

To measure investors' pre-existing knowledge of management quality, we collect data on newspaper coverage of NNRs from *WiseNews*, a leading Chinese newspaper aggregator. We measure the degree of information availability by whether any NNR was covered by the top 10 newspapers (by average circulation) in 2016.³⁴ Column (1) and (2) of Table 8 report the results. We find that NNR municipalities with different levels of information asymmetry associated with their own NNRs experience similar increases in MCB spreads around GSA, compared to non-NNR municipalities.

Next, we test whether the effect of management quality differ depending on pre-existing news coverage. Our simple conceptual framework implies that areas with poorly managed NNRs and no news coverage of these violations ex-ante should experience both the regulatory cost effect and the preferences effect. On the other hand, areas with poorly managed NNRs that have been covered by the news ex-ante will only experience the regulatory cost effect. To test these implications in the data we estimate the heterogeneous effects of ex-ante illegal activities within NNRs separately in the samples with high vs low news coverage. The results are reported in columns 3 and 4 of Table 8. We find point estimates on the heterogeneous effects of ex-ante illegal activities that are larger in areas with low news coverage, where both effects should be at play. However, the difference is

³³ As standard in the triple-difference approach, we include the triple interaction term, three dual interaction terms, and three single terms in the regression in Equation (1) and all heterogeneity tests, with some potentially absorbed by fixed effects.

³⁴ These include: People's Daily (ren min ri bao), Reference News (can kao xiao xi), Southern Metropolis Daily (nan fang du shi bao), Xinhua Daily Telegraph (xin hua mei ri dian xun), Qianjiang Evening News (qian jiang wan bao), Peninsula Metropolis Daily (ban dao du shi bao), Metropolitan Express (du shi kuai bao), Guangzhou Daily (guang zhou ri bao), Guangzhou Daily Community Edition (guang zhou ri bao she qu ban), Qilu Evening News (qi lu wan bao).

not statistically significant, as shown in column (5) where we test this more formally.

Taken together, the evidence in this section supports a regulatory cost interpretation of GSA’s impact. Effects are concentrated in poorly-managed NNRs where compliance costs are highest, yet do not vary meaningfully with prior information availability. The absence of heterogeneity by ex-ante media coverage of NNRs is consistent with two interpretations: investors already understood management quality differences, and/or investors with non-pecuniary biodiversity preferences are not marginal in this market. Both explanations imply that our estimates mostly capture the financing cost of regulatory enforcement, i.e. the expected fiscal burden of compliance rather than punishment from biodiversity-conscious investors. This interpretation aligns with our broader finding in [Section 6](#) that investors do not reward municipalities achieving greater biodiversity improvements, suggesting limited scope for non-pecuniary preferences in this market.

5.3 ALTERNATIVE EXPLANATIONS FOR PUBLIC FINANCING ACTIVITIES

Increased demand for financing by local governments. An alternative channel through which GSA could contribute to the widening of the MCB spreads is by exerting greater upward pressure on the demand curve of NNR municipalities. We examine the impact of GSA on the amount of local public financing based on the specification in [Table 5](#). [Table A2](#) reports the effects on (1) the probability of new MCB issuance, (2) the amount of new MCB issuance, (3) the growth rate of total local public debts, (4) the growth rate of local public interest-bearing debt. The latter two metrics are measured in a way consistent with the methodology in [Section 5.1](#). We find noisy and non-significant effects on outcomes capturing new local public borrowing. This result suggests that even though local governments’ demand for funds might grow after GSA, there was a constrained willingness of MCB investors to provide capital due to concerns about the creditworthiness of the NNR municipalities. The latter effect might have played a role in determining the amount of financing available to these municipalities, driving up the MCB risk premium.

Local political risk. As outlined in [Section 2.2](#), GSA holds officials accountable for negligence, which might affect the political turnover (the ex-post perspective) and change the promotion incentive for officials (the ex-ante perspective). For instance, in the “Qilian Mountain Incident,” the area under the jurisdiction of Zhangye City recorded the most violations. Following the central inspection, Shengwu Mao, the former municipal secretary of the CPC in Zhangye, and Zeyuan Huang, the incumbent mayor of Zhangye, received a severe warning within the party, which primarily meant they would not be allowed to be promoted for a minimum period of one and a half years. It is also worth noting that

the punishment one level higher than a severe warning leads to dismissal.

From some MCB investors' perspective, GSA adds evaluation requirements for local officials, which possibly creates additional political risk. Thus, the perceived rise in risk premium might be attributed to the market's concern about the stability of the local political environment. To explore this mechanism, we first study changes in the positions of the municipal secretary of the CPC and the mayor, two primary leaders in Chinese city governments. Again, using the city-year-level specification in Table 5, we find no significant effects of GSA on local political turnover (Panel A of Table A3). We also employ a triple difference specification to study whether the impact of GSA is different during the first two complete years of the leading official's term. Prior studies reveal that officials in the early stages of their term (i.e., the first two years) tend to have weaker political incentives, compared with their more established counterparts (Buntaine *et al.*, 2024). Intuitively, during the late stage of their term, officials have limited leeway to incur penalties such as severe warnings that influence their promotion prospects for several years. As presented in Panel B of Table A3, we do not observe significant heterogeneity across different stages of officials' tenures. Overall, these results suggest that the potential increase in local political risk is unlikely to be driving the pricing effect of GSA in the MCB market.

6 BIODIVERSITY IMPROVEMENT, INVESTOR VALUATION, AND POLICY

The preceding discussion has focused on the negative financial impact of GSA. How about its biodiversity impact? Has GSA merely raised concerns or genuinely improved local biodiversity as intended by Beijing? Either way, do bond investors think about biodiversity at all? We discuss and empirically investigate these questions in what follows.

6.1 IMPACT OF GSA ON BIODIVERSITY

Because the observation of biological systems is more complicated than physical systems, there have been no reliable and widely accepted metrics to measure the performance of biodiversity finance (Karolyi & Tobin-de la Puente, 2023). Nor are there currently complete panel statistics on various species at a fine geographical level in China. As a proxy for biodiversity, we focus on examining changes occurring in a particular animal group – birds – as we can access two specific datasets that capture the dynamics of bird species in China. Birds are an important indicator group for biodiversity due to their wide-ranging habitat and sensitivity to the environment.

Our primary dataset is from the China Bird Report Center (CBRC), the leading Internet bird data recording and sharing platform. The data structure of the CBRC

resembles that of the eBird Reference Dataset, a citizen science dataset containing reports from users with detailed descriptions of their birdwatching trips, as well as the species of birds observed. We construct panel data of birdwatching activities at the city-quarter level from 2015 to 2021, using reports uploaded to the CBRC. We exclude the city-quarter observations with no birdwatching report, and then empirically explore the effects of GSA on the number of bird species observed through our baseline specification.

The results are reported in [Table 9](#). As shown in Column (1), following GSA, more bird species were reported in NNR municipalities compared to non-NNR municipalities. Although the records from voluntary observations cannot convincingly represent the actual status of local wildlife, it seems reasonable that the more abundant the bird population, the more likely it is to be observed. Thus, the results are at least consistent with the notion that local biodiversity benefited from conservation actions taken by the government through the enforcement of GSA. Furthermore, Columns (2) and (3) show that the intensity of birdwatching activities had no significant change around GSA, evidenced by the number of reports and reporters. This alleviates the plausible concern that the increase in bird species observed could be driven by more attention from birdwatchers in NNR municipalities compared to that in non-NNR municipalities after GSA.

We also use data from the bird monitoring information system of the Chinese National Ecosystem Research Network (CNERN), an information-sharing platform constructed by the Ministry of Science and Technology, integrating resources of multiple existing field observation and research stations. Specifically, the data on birds are summarized from eleven forest ecological stations that conduct a systematic survey of birds every five years in the corresponding areas.³⁵ Since these stations are all located in NNR municipalities, we cannot employ a DID estimation and only perform descriptive statistics at the time series level. [Figure A2](#) shows that the number of bird species surveyed declined substantially in the first fifteen years of this century, but recovered following GSA. Although it is only suggestive evidence, this pattern is consistent with the poor management of NNRs before GSA and some real positive effects of this reform on biodiversity.

In addition to bird watching data, we attempt to measure the impact of GSA on local biodiversity using the International Union for Conservation of Nature’s Red List of Threatened Species (IUCN Red List) database, a leading source on global species extinction risk. This database includes information on over 163,000 species, with spatial data available for 83% of such species starting from 2017 ([IUCN, 2022](#)). We use the IUCN Red List data to construct a measure of number of “potentially observed” species in each location for the years 2017 and 2022. For each Chinese city, we calculate the average

³⁵The stations are: Ailao Mountain Station, Beijing Forest Station, Banna Station, Changbai Mountain Station, Dinghu Mountain Station, Gongga Mountain Station, Heshan Station, Huitong Station, Maoxian Station, Qingyuan Station, Shennongjia Station.

number of species across the grid cells within the city’s administrative boundaries. The number of species include amphibian, bird, and mammal species. Notice that we only have information on number of species from the IUCN in 2017, which we consider as pre-GSA period, and 2022, which we consider as post-GSA period.

The results are reported in [Table A4](#). We find that NNR municipalities experience a significantly larger increase in the number of potentially observable species between 2017 and 2022, consistent with positive real effects of GSA on local biodiversity. The magnitude of the estimated coefficient in Column (1) indicates a 4.6% larger increase in number of species in NNR municipalities relative to the average number observed in non-NNR municipalities in the baseline year 2017 (260 species). In Column (2) of [Table A4](#) we use as outcome variable the city ranking in number of species within China (where a lower number indicates a higher ranking). As shown, NNR municipalities gain on average 10 positions in the ranking of Chinese cities based on number of species.

6.2 INVESTORS’ CONSIDERATION FOR BIODIVERSITY IMPROVEMENT

We now investigate a countervailing force exerted by investors with biodiversity preferences, including non-pecuniary considerations. They may reward NNR municipalities that demonstrate effective ecological improvements. This mechanism is crucial for rethinking how sustainable finance works. Although the net effect of GSA reveals a prevailing market apprehension about the costs incurred by ecological transition, favorable investment strategies that focus on biodiversity improvements as a declared objective could mitigate the increase in the public cost of capital, thereby benefiting social welfare.

To test this channel, we construct a municipality-level proxy of observed biodiversity improvement around the introduction of GSA. We categorize the birdwatching data introduced in [Section 6.1](#) into two phases: from January 2015 to June 2017, and from July 2017 to December 2021. For each NNR municipality, we aggregate the number of bird species reported, subsequently calculating the corresponding changes from the pre-GSA to the post-GSA period. Then, we compare the effect of GSA on bond spreads in the top quartile of NNR municipalities, which exhibit a substantial increase in bird species observed around GSA, with other NNR municipalities. To enhance robustness, we use changes in the number of species from 2017 to 2022, based on the IUCN Red List data described in [Section 6.1](#), as an alternative measure of biodiversity improvements. The results presented in [Table A5](#) reveal that these NNR municipalities with greater improvements in local biodiversity did not experience a lower rise in MCB spreads. The role of pricing conservation costs behind biodiversity enrichment seems to dominate.

In addition, cross-sectional tests on ex-ante information asymmetry associated with NNRs discussed above also indicate an insignificant impact of incentives to reward efforts

on protecting biodiversity. Specifically, if NNRs with higher biodiversity value are more likely to be covered by newspapers, then NNR municipalities with more NNR news should have access to more favorable financing. This assumption implies the same sign of heterogeneity across municipalities with different levels of NNR news coverage as that under the hypotheses of punishment for dereliction and concern for biodiversity loss. However, [Table 8](#) shows very small differences associated with ex-ante media attention on NNRs.³⁶

6.3 AGGREGATE FINANCING COSTS

We propose a back-of-the-envelope calculation of the additional financing costs for LGFVs with outstanding MCBs, as a way to assess the aggregate financing costs of GSA. Specifically, we compute the interest-bearing debts of LGFVs and, subsequently, the rises in interest payments caused by GSA. We examine how these additional financing costs changed from 2013 to 2021 and how much GSA contributed to this change.

As depicted in [Figure 4](#), the debt cost escalated following the GSA implementation: public financing costs (the solid red line) began to rise substantially in 2018 and reached approximately 120 billion US dollars in 2021.

To compute the counterfactual debt servicing costs for Chinese municipalities in the absence of GSA we proceed as follows. First, we estimate a version of [Equation \(2\)](#) in which observations are weighted by the initial level of outstanding debt in each municipality.³⁷ Weighting municipalities by their debt size allows us to obtain estimates of the effect of NNRs on spreads that are representative of the aggregate Chinese economy rather than of the average municipality in our sample. Next, for each NNR municipality and year, we compute the counterfactual spread they would have faced in the absence of an NNR by subtracting the estimated coefficients in the first step from the observed spread. Finally, we multiply the counterfactual spread by the level of debt of each municipality and sum across all municipalities to obtain the aggregate counterfactual financial cost.

Counterfactual financing costs in the absence of GSA are represented by the dashed blue line in [Figure 4](#). The gap between actual and counterfactual financing costs has been

³⁶While theoretical arguments suggest that investor type heterogeneity in bondholdings could provide evidence of biodiversity prioritization, limitations of publicly available data prevent a comprehensive analysis of all investor transactions. Nonetheless, Chinese regulations mandate quarterly disclosure of their top five or top ten bond holdings by public mutual funds. As shown in [Table A6](#), the GSA’s impact does not significantly differ between bonds previously among the top holdings of mutual funds and other bonds. This lack of a differential impact, from an investor type perspective, further implies that the MCB market may indeed lack strong support for biodiversity values.

³⁷Specifically, the city-level average value of LGFV interest-bearing debts from 2013 to 2016. We also conduct a robustness test using the weighted least square (WLS) approach to estimate [Equation \(1\)](#) and still observe a statistically significant increasing effect of GSA on MCB spreads (see [Table A7](#)).

increasing since the introduction of GSA. In total, the additional LGFV interest payments incurred by GSA from 2018 to 2021 amounted to approximately 40 billion dollars.

Recent estimates by [Deutz *et al.* \(2020\)](#) indicate that the gap between the funding needs and the actual funding available to preserve biodiversity at the global level (or biodiversity financing gap) is approximately 711 billion dollars per year. If countries were to share such biodiversity spending in proportion to their share of GDP, China would need to spend 131.54 billion dollars per year on protecting biodiversity (see [Table A8](#)). However, these estimates do not take into account the extra financing costs associated with biodiversity conservation, which in the case of GSA in China accounts for 4.21%, 8.75%, and 16.24% of the total costs (131.54 billion dollars) in 2019, 2020, and 2021, respectively.³⁸ This suggests that, in practice, the cost of protecting biodiversity is higher than what is implied by simply considering the direct investment in conservation. Empirical evidence from GSA indicates that indirect costs related to changes in the cost of capital can be of significant magnitude, and scholars and policymakers should not overlook them in designing global biodiversity conservation frameworks.

7 ADDITIONAL RESULTS AND ROBUSTNESS TESTS

7.1 ALTERNATIVE SPECIFICATIONS

We test the robustness of the baseline estimates to a range of alternative specifications. First, we examine alternative measures of treatment intensity. Our baseline specification uses a binary treatment variable. In this section, we use the proportion of NNRs within the boundaries of a municipality, which captures continuous variations in pressure on municipalities to manage NNRs. Panel A of [Figure A3](#) reports the results obtained by replacing the binary variable with alternative measures. Using more flexible measures of exposure to GSA does not alter the baseline findings.

Second, in Panel B-H of [Figure A3](#), we report the results of alternative measures of bond spreads. Specifically, we utilize the quarterly median of spreads, the quarterly mean of spreads, quarterly trading-volume-weighted average spreads, and measures of spread obtained replacing the risk-free benchmark with China’s Treasury yields. All results are robust to using these alternative measures of the outcome.

Third, we examine the robustness of the main results to using alternative data frequency for MCB trades and the bond issuance data. The corresponding sampling processes are the same as the quarterly secondary market datasets in the baseline specifica-

³⁸We also calculated the biodiversity financing gap allocated to China based on its share of global species (land area). We find that the additional interest payments brought by GSA as a percentage of the biodiversity investment demand for the respective years 2019 - 2021 were 12.18% (11.13%), 25.30% (23.13%), and 46.97% (42.94%).

tion. The results are reported in [Figure A4](#) in the order of monthly frequency, semi-annual frequency, annual frequency, and the offering sample. Estimates remain significant regardless of the data frequency used. In addition, the results remain robust to excluding specific sub-samples of municipalities or alternative definitions of the control group.³⁹

7.2 CONTEMPORANEOUS POLICY SHOCKS

We also investigate whether other events related to environmental issues occurring during the same period could affect the MCB market. Two most prominent policies initiated by the central government around the introduction of GSA are: (1) the “Central Inspection on Environmental Protection” and (2) the “Nationwide Campaign to Prevent and Control Pollution.” We describe the institutional background next.

Central Inspection on Environmental Protection (CIEP). This initiative was approved by the central government in July 2015 and launched in 2016, and had the objective to inspect and improve environmental conditions in each province.⁴⁰ An essential aspect of this initiative was the establishment of a new mechanism for environmental inspection: specific central inspection groups were dispatched to check local environmental protection and policy implementation on a provincial basis. Two rounds of these inspections have been conducted so far: the first round lasted from January 2016 to September 2017, while the second one took place from July 2019 to June 2022.

Nationwide Battle to Prevent and Control Pollution (NBPCP). In June 2018, Beijing announced a comprehensive plan for pollution reduction, named Nationwide Battle to Prevent and Control Pollution.⁴¹ The plan aimed at improving the country’s environmental quality by reducing the emissions of major pollutants. The plan targeted pollution in air, water, and land, setting specific goals to be achieved by the end of 2020. More specifically, the proportion of excellent air quality days in cities should reach over 80%; sulfur dioxide emissions should be reduced by at least 15% compared with the level in 2015; the proportion of surface water bodies classified as Grade I-III should exceed 70%;

³⁹[Table A9](#) reports these additional robustness tests. It shows that the main results are robust to: (i) excluding NNR municipalities in the top quartile of the distribution of *NNR area* in the country, (ii) only using non-NNR municipalities adjacent to an NNR municipality as control group, (iii) only using non-NNR municipalities within a 50 km radius from an NNR municipality as control group, and (iv) using the intersection set of samples from restrictions (i) to (iii).

⁴⁰The Ministry of Ecology and Environment of the CPC has released detailed information on the Central Inspection on Environmental Protection (see: <https://www.mee.gov.cn/ywgz/zysthjbhdc/>).

⁴¹The corresponding policy document is the Guidelines on Comprehensively Strengthening Ecological and Environmental Protection and Resolutely Fighting the Battle to Prevent and Control Pollution, June 2018, (see: https://www.gov.cn/zhengce/2018-06/24/content_5300953.htm?eqid=804df71900054d45000000056461879e).

the proportion of nearshore seawater classified as excellent (Grade I, II) should surpass 70%; the safe utilization rate of polluted plots should be over 90%.

The timing of these policies partly overlaps with the launch of GSA in the second half of 2017. All these policies target environmental or ecological issues. CIEP and NBPCP primarily address environmental risks and regulate private pollution activities, while GSA focuses on local government contributions to biodiversity, particularly in NNRs.

To disentangle the impact of GSA from other contemporaneous and related policies, we construct measures of exposure to such policies and include them as controls in our estimation. In [Table A10](#), we measure the impact of CIEP by using four indicators: (1) *In the 1st round*, a dummy variable that equals one for province-quarters in the first round of CIEP investigations; (2) *In the 2nd round*, a dummy variable that equals one for province-quarters in the second round of CIEP investigations; (3) *After the 1st round*, a dummy variable that equals one for provinces investigated in the first round after the first quarter of investigation; (4) *After the 2nd round* a dummy variable that equals one for provinces investigated in the second round after the first quarter of investigation. Overall, we find that after the first round of CIEP, MCB spreads rise more for bonds originating in investigated provinces. However, the coefficients capturing the impact of GSA remain stable in magnitude and statistically significant.

[Table A11](#) includes measures of exposure to NBPCP. Considering that NBPCP has quantifiable targets for pollution reduction and primarily focuses on enterprise pollution behaviors, we select four city-year-level pollution metrics in 2017: the Air Quality Index, industrial sulfur dioxide emissions, industrial wastewater emissions, and industrial dust emissions. The Air Quality Index, as disclosed by the Ministry of Ecology and Environment, is a composite index weighted by the concentrations of six major air pollutants, with higher scores indicating more severe local air pollution. The other three metrics, collected from the local Bureaus of Statistics, are scaled by the GDP of the secondary sector. We assume that the higher the pollution level, the greater the price that a city needs to pay to comply with NBPCP. We interact these metrics with $PostNBPCP_t$, a dummy variable that equals one if the first quarter of 2018 and zero otherwise, and then add interaction terms into our baseline specification. The results indicate that initially more polluted cities (as measured by sulfur dioxide emissions in particular) indeed experienced a higher increase in bond yields after the launch of NBPCP. Still, this seems an independent effect from that of GSA, as the coefficients capturing the impact of GSA remain stable in magnitude and statistically significant.

Macro dynamics of the Chinese bond market. NNR municipalities could also have been more exposed to other contemporaneous shocks facing the Chinese credit and bond

market which underwent several regulatory changes during our sample period following an increase in default events. These changes included a forceful push for a more efficient bankruptcy regime (Li & Ponticelli, 2022), and an intensified scrutiny of risky financing activities, in particular the 2018 New Regulations on Asset Management introduced to contain the growth of the shadow banking industry. As documented in Geng & Pan (2024), the New Regulations led to a substantial increase in the SOE premium – i.e. the difference in spreads between bonds issued by non-SOE and SOE.

We think that these events, are unlikely to have a significant confounding effects. First, all bonds in our sample are issued by local governments, which are SOEs. Second, we conduct a robustness test in which we restrict our analysis to MCBs with AAA-rating, the top grade in the Chinese bond market, for which default was not a concern. The working assumption of this exercise is that bonds with the same top-notch rating should be similarly affected by reforms aimed at reducing systemic risk in the bond market. This robustness test is reported in Table A12. The results obtained with the sub-sample of AAA-rated bonds are smaller in magnitude relative to the full-sample results (0.18 vs 0.23), but remain close to 20 basis points and statistically significant.

Third, we conduct an additional placebo test in which we examine changes in bond yield spreads for a non-MCB sample around GSA. Specifically, we focus on centrally state-owned enterprises (CSOEs) across different regions, which, compared to other locally state-owned enterprises and private enterprises, are less dependent on local government fiscal conditions. As a result, their bond yield spreads more directly reflect local economic and financial fundamentals, as well as macroeconomic fluctuations related to the bond market. Table A13 shows that CSOE bond spreads in NNR municipalities did not exhibit significant differences compared to those in non-NNR municipalities following the GSA implementation. This finding further addresses concerns about confounding factors in market dynamics and local development beyond exposure to GSA.

7.3 LOCAL PUBLIC DEBT BURDEN

We also utilize cross-sectional variations in the pressure for local public debt repayment to explore the market’s expectations of the default probability for MCBs. Theoretically, given the transition costs to comply with GSA, cities with higher public debt burdens are more likely to face financial distress and challenges in servicing their bonds. Conversely, cities with low public debt burdens may still possess sufficient funds to safeguard the bondholders’ interests. Therefore, in terms of pricing, MCB spreads of the former regions would increase more than those of the latter.

We construct two measures of the local public debt burden. In China, all entities requesting authorization to issue a corporate bond in a given year are required to disclose

their liabilities for the current year and (at least) the three previous years. In addition, during the outstanding period of MCBs, the issuer should also publicly reveal its liability terms on a regular basis. Following [Huang *et al.* \(2020\)](#), we perform a conservative calculation of the debt of LGFVs with bond outstanding, and aggregate LGFVs’ total debts and interest-bearing debts to the city-year level. The city-year aggregate measures are then divided by the corresponding city’s GDP in that year. We define a dummy variable, *High debt burden*, which equals one if a city’s level of public debt burden is in the top quartile of the distribution for the year before the bond trade and zero otherwise.

[Table A14](#) presents the estimates from fully interacted models which include the interaction of each of the two measures of *High debt burden* with our baseline treatment indicator. The coefficients of the triple interaction term are positive, precisely estimated, and larger in magnitude than the interaction between NNR municipalities and the Post-GSA dummy. This indicates that the effect of GSA on MCB spreads is significantly larger for cities with high initial levels of debt burden. This result further corroborates the notion that there is a concern about the heightened risk of a MCB default from MCB investors following GSA.

7.4 BOND MATURITY

The recovery of biodiversity within NNRs and the generation of its sustained social benefits are likely to necessitate a long-time horizon. However, the transition costs associated with GSA already emerged in the short run. Thus, if the additional increase in public financing costs in NNR municipalities was driven by investors’ concerns over transition costs, the pricing effects should be more pronounced on short-term MCBs.

To explore this heterogeneity in the bond term structure, we examine the effects of GSA on MCB spreads across different levels of bond time to maturity. The term structure cutoff is set as 3 years or 4 years depending on the test. As described in [Section 3.2](#), the maturity of credit bonds issued in the Chinese market is shorter than that in the U.S. market. In our sample, both the mean and median of residual maturity of MCB observations fall within the range of 3 and 4 years. [Table A15](#) shows that both long- and short-term MCBs are impacted by GSA, although the effect is larger for short-term bonds.

8 CONCLUSION

We study the relationship between biodiversity conservation and its cost imposed on the public financing of governments. In particular, we exploit the Green Shield Action (GSA) in China as a shock to local financial and fiscal burdens and study market re-

sponses in the municipal corporate bond (MCB) markets. Evidence indicates that cities with National Nature Reserves (NNRs) experienced greater increases in the general public borrowing costs after the implementation of GSA. The effects come from the transition pressure from pre-existing economic activities within NNRs and increased public spending on biodiversity following the policy. These constraints and costs worsen local fiscal conditions, amplifying investors' concern about the probability of local government default.

Our findings cannot be explained by the financing demand of local governments, heightened political risk, or related information disclosure. We also find that, although GSA improved biodiversity, MCB investors are not considering these ecological values when making decisions, nor are they financially punishing cities revealed with more ecological degradation. Finally, we provide a conservative estimate of the additional public financing costs GSA has incurred. Given China's abundant natural capital, economic scale, and representative biodiversity, as well as the global nature of biodiversity challenges, our findings provide valuable benchmarks for future research and practice.

Our work likely has several policy implications applicable in general settings. First, estimating the financing gap for biodiversity conservation requires considerations beyond direct investments for biodiversity transition. It needs to account for any additional costs in the financial market. Thus, our study is informative given the current global acceleration of biodiversity loss and the urgent need for collective efforts to save natural capital. Second, we provide insights into how investors' lack of internalization of the (long-run) environmental benefits of government policies may hinder sustainable development goals. Policymakers must consider the reactions of financial markets when formulating relevant policies for sustainable development. More detailed disclosure, dedicated financing vehicles, and promotions for the social recognition of biodiversity value constitute key steps in reversing the trend of nature capital loss.

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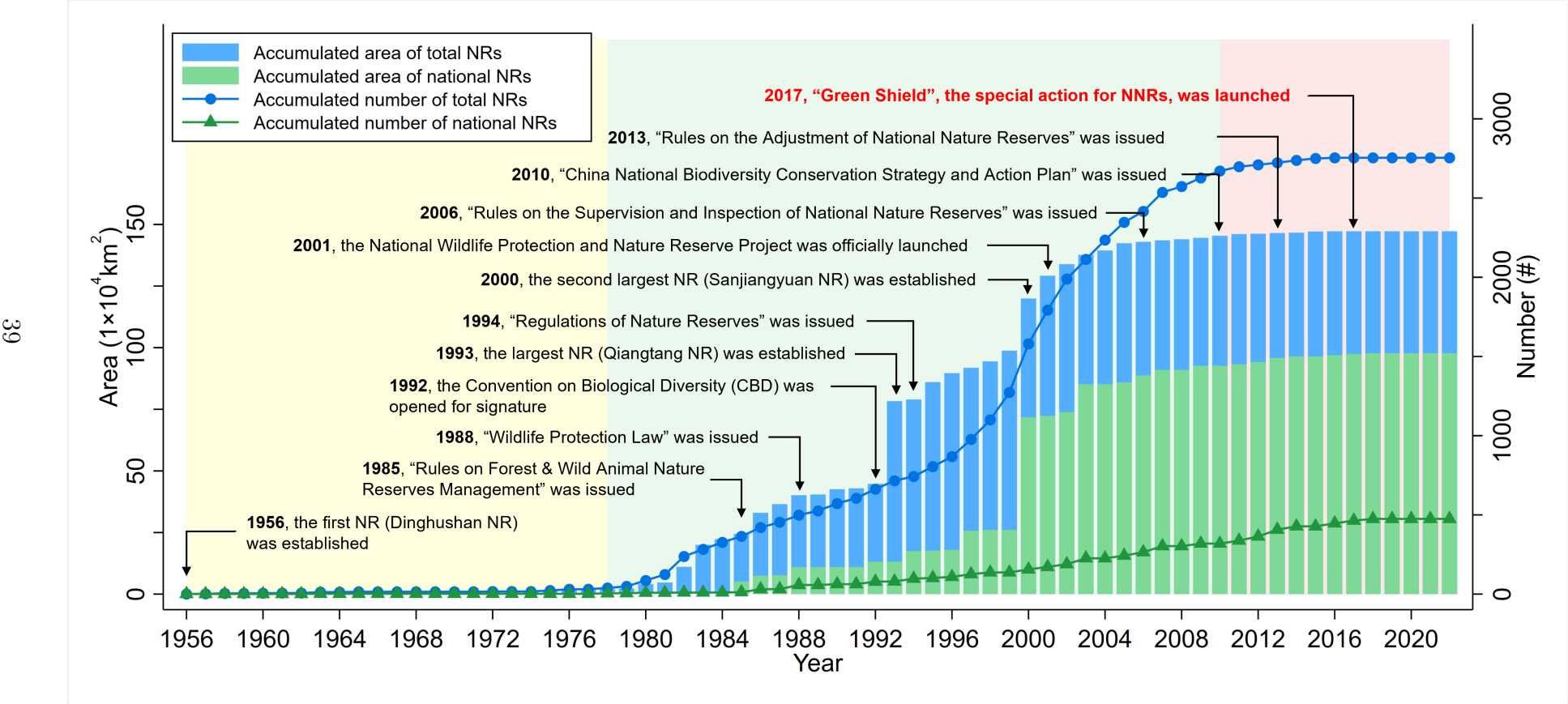
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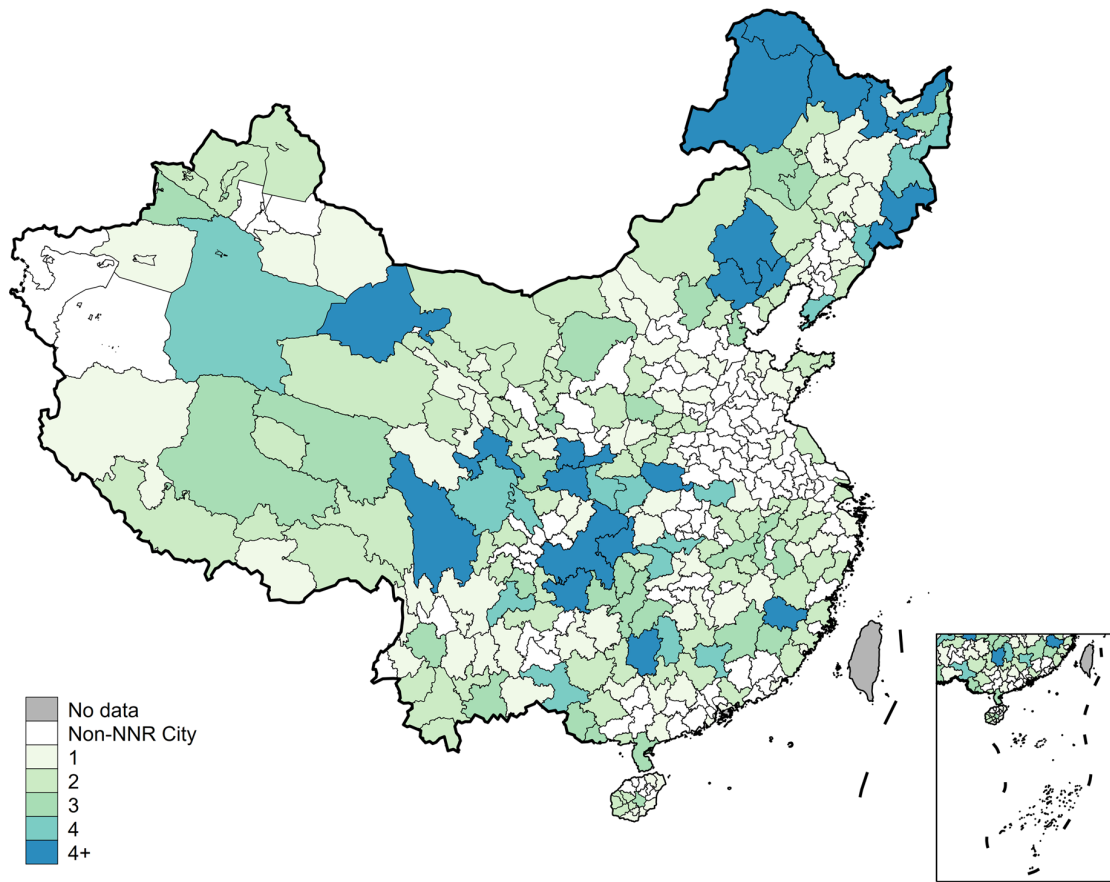
FIGURES AND TABLES

FIGURE 1: DEVELOPMENT OF NATURE RESERVES AND IMPORTANT RELEVANT EVENTS IN CHINA



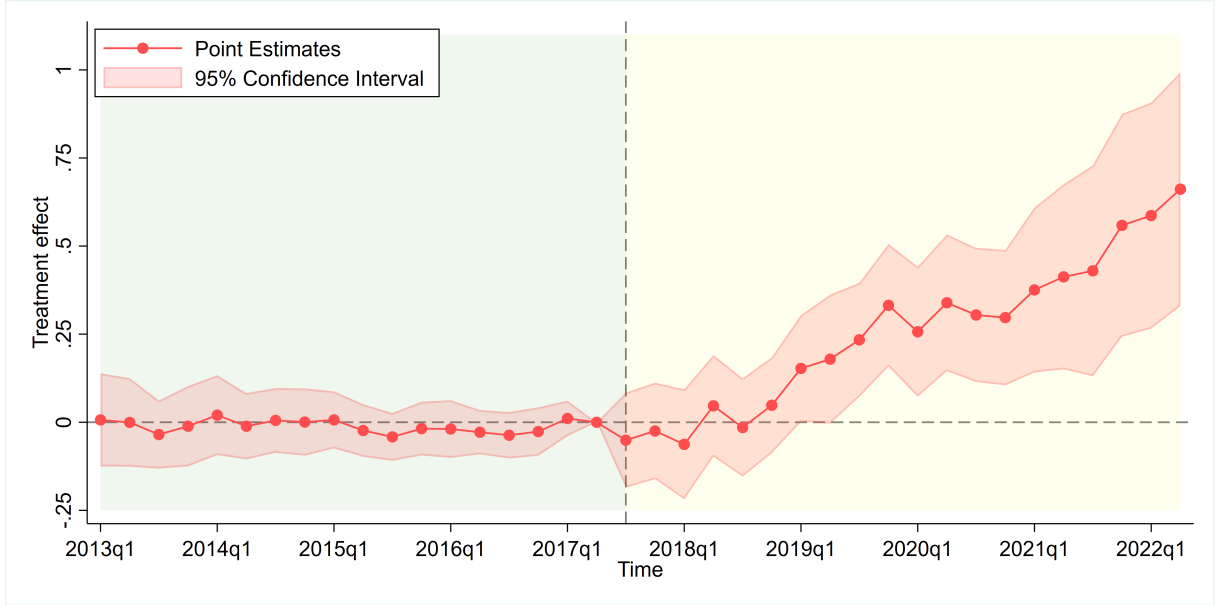
Notes: This figure illustrates the number (right axis) and area (left axis) of National Nature Reserves (NNRs) in China from 1956 to 2022. Important events are also marked with arrows in the corresponding year.

FIGURE 2: THE GEOGRAPHICAL DISTRIBUTION OF NNRs AT THE CITY LEVEL



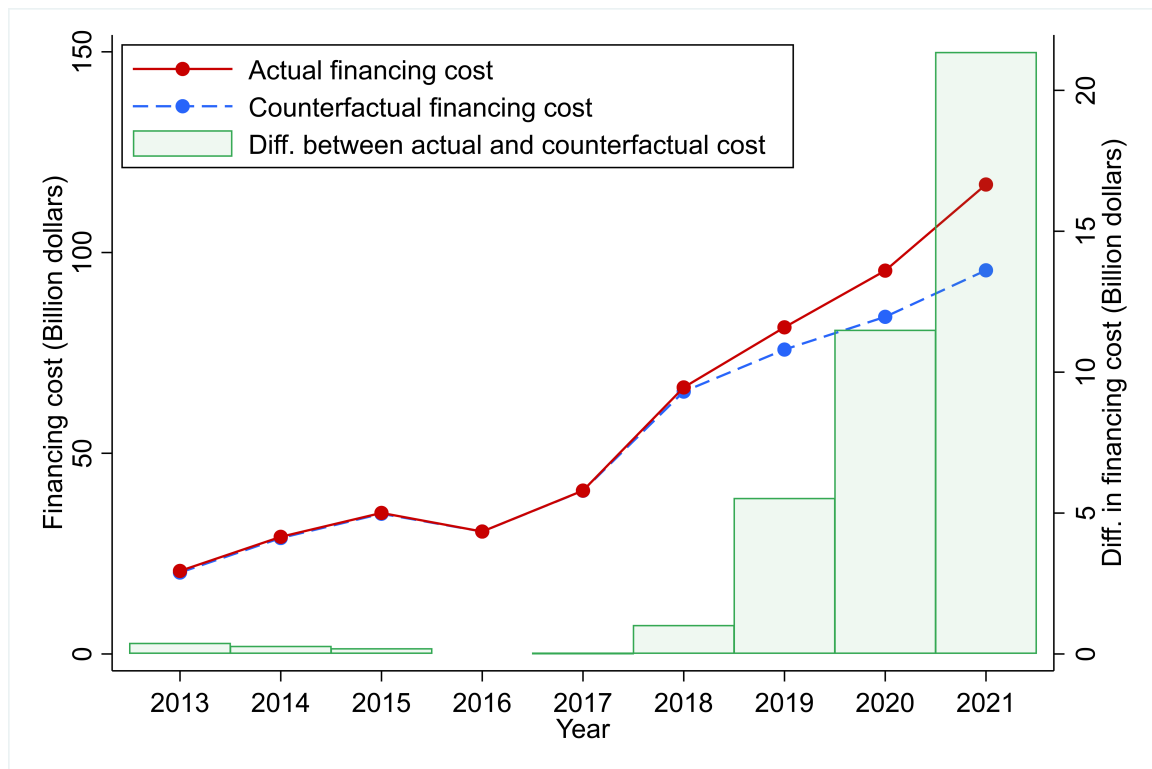
Notes: This figure illustrates the number of National Nature Reserves (NNRs) in each city in China as of 2016. The regional boundaries are delineated down to the city level. The legend depicts the range of the number of NNRs corresponding to each color depth.

FIGURE 3: CHANGES IN MCB SPREADS BEFORE AND AFTER GSA



Notes: This figure depicts the estimated differences in MCB spreads between NNR municipalities and non-NNR municipalities before and after GSA. The markers and capped spikes represent the OLS estimators and 95% confidence intervals. The dependent variable is the MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. The dashed vertical line represents the time when GSA was launched. The reference is the second quarter of 2017. The regression follows [Equation \(2\)](#). The standard errors are clustered at the city level.

FIGURE 4: LOCAL PUBLIC DEBT COST COMPARISON OF TRUE VALUE AND COUNTERFACTUAL ESTIMATES



Notes: This figure plots the aggregate cost of LGFVs' debt service over time with and without GSA. The solid red line represents the actual aggregate cost of LGFVs' debt service (the left axis). The blue dashed line represents the counterfactual aggregate cost of LGFVs' debt service (the left axis). The green bar represents the difference between the actual and counterfactual cost (the right axis) over time. The exchange rate between the US dollar and the CNY is set as 1: 7.

TABLE 1: SUMMARY STATISTICS

	Obs	Mean	St.D	P25	P50	P75
<i>Outcome:</i>						
Spread	87885	1.958	1.283	1.014	1.664	2.509
<i>Treatments:</i>						
NNR	87885	0.511	0.500	0.000	1.000	1.000
Post	87885	0.626	0.484	0.000	1.000	1.000
<i>Bond characteristics:</i>						
ln(Bond size)	87885	2.225	0.530	1.946	2.303	2.565
Time to maturity	87885	3.834	1.757	2.364	3.680	5.148
Option	87885	0.692	0.462	0.000	1.000	1.000
Guarantee	87885	0.230	0.421	0.000	0.000	0.000
Exchange	87885	0.323	0.468	0.000	0.000	1.000
Bond rating	87885	2.290	0.822	2.000	2.000	3.000
Issuer rating	87885	2.654	0.732	2.000	3.000	3.000
<i>City pre-shock features:</i>						
ln(GDP)	87885	3.433	0.948	2.669	3.476	4.234
ln(GDP per capita)	87885	3.982	0.511	3.707	4.051	4.412
GDP growth rate	87885	10.493	1.612	9.400	10.600	11.771
Tertiary sector in GDP	87885	42.938	9.061	36.489	41.979	47.800
ln(Population)	87885	1.745	0.673	1.354	1.718	2.163
Population growth rate	87885	0.705	1.105	0.114	0.416	1.029
ln(Nighttime lights)	87885	2.488	0.753	1.928	2.623	3.038
ln(Fixed investments)	87885	3.063	0.813	2.497	3.008	3.753
ln(Housing price)	87885	1.783	0.441	1.440	1.717	2.043

Notes: This table reports summary statistics in the baseline sample. The main dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Bond controls contain (ln)bond issuing amount, bond maturity, bond rating, issuer rating, and whether the bond is option-embedded, guaranteed, and traded on exchange. City pre-shock variables include (ln)GDP, (ln)GDP per capita, GDP growth rate, the proportion of tertiary sector in GDP, (ln)population, population growth rate, (ln)nighttime light intensity, (ln)fixed investment, and (ln)housing price at the city level in 2013. The dependent variable and city-level continuous variables are winsorized at the 1st and 99th percentiles. Detailed variable definitions are presented in [Table A1](#).

TABLE 2: PRE-GSA LEVELS AND TRENDS IN MUNICIPALITY CHARACTERISTICS

	NNR=1	NNR=0	Difference
	(1)	(2)	(1)-(2)
ln GDP (ten billion RMB)	2.527 (0.066)	2.669 (0.089)	-0.142 [0.109]
Δ ln GDP (ten billion RMB)	0.161 (0.014)	0.177 (0.016)	-0.016 [0.021]
ln GDP per capita (thousand RMB)	3.648 (0.042)	3.683 (0.052)	-0.035 [0.067]
Δ ln GDP per capita (thousand RMB)	0.141 (0.013)	0.158 (0.015)	-0.017 [0.021]
GDP annual growth rate (%)	10.245 (0.163)	10.763 (0.196)	-0.518** [0.257]
Δ GDP annual growth rate (%)	-3.194 (0.258)	-3.477 (0.301)	0.283 [0.402]
Tertiary sector GDP (%)	37.657 (0.715)	38.375 (0.811)	-0.718 [1.106]
Δ Tertiary sector GDP (%)	6.335 (0.345)	6.277 (0.408)	0.058 [0.542]
ln Nighttime light intensity	1.972 (0.054)	2.042 (0.072)	-0.070 [0.089]
Δ ln Nighttime light intensity	0.089 (0.005)	0.065 (0.005)	0.024*** [0.007]
ln Housing price (thousand RMB/ m^2)	1.488 (0.030)	1.528 (0.033)	-0.040 [0.046]
Δ ln Housing price (thousand RMB/ m^2)	0.090 (0.011)	0.090 (0.013)	-0.000 [0.018]
ln Fixed investment (ten billion RMB)	2.290 (0.058)	2.371 (0.078)	-0.081 [0.095]
Δ ln Fixed investment (ten billion RMB)	0.256 (0.033)	0.273 (0.039)	-0.017 [0.052]
ln Population (million)	1.171 (0.055)	1.280 (0.069)	-0.110 [0.088]
Δ ln Population (million)	0.006 (0.004)	0.018 (0.004)	-0.012** [0.006]
Population annual growth rate (%)	0.286 (0.096)	0.482 (0.112)	-0.197 [0.150]
Δ Population annual growth rate (%)	0.062 (0.095)	0.212 (0.136)	-0.151 [0.161]
Urbanization rate (%)	30.150 (1.094)	28.972 (1.373)	1.178 [1.755]
High school and above education rate (%)	21.804 (0.570)	21.749 (0.821)	0.055 [0.970]
ln Local fiscal revenue (million RMB)	2.296 (0.074)	2.413 (0.095)	-0.116 [0.120]
Δ ln Local fiscal revenue (million RMB)	0.163 (0.018)	0.180 (0.023)	-0.017 [0.029]
ln Local fiscal expenditure (million RMB)	3.202 (0.049)	3.142 (0.061)	0.060 [0.079]
Δ ln Local fiscal expenditure (million RMB)	0.281 (0.012)	0.293 (0.015)	-0.012 [0.019]

Notes: This table presents pre-GSA economic characteristics for NNR and non-NNR municipalities (Columns (1) and (2), respectively) and their differences (Column (3)). Urbanization and high school education rates are from the 2010 census (no suitable observations are available to calculate pre-GSA trends due to the decennial nature of the population census in China). The other variables in levels are each measured as of 2013 (the start of our sample period), and their trends (marked with Δ) compare the data from 2016 (the year just before the GSA launch) to that of 2013. Standard deviations are in parentheses and standard errors are in brackets. Detailed variable definitions are presented in [Table A1](#). Cities are restricted to those in baseline sample (consistent with [Table 1](#) and [Table 3](#)). The exchange rate between the US dollar and the CNY was typically between 1:6 and 1:7.2.

TABLE 3: GSA AND MCB SPREADS: BASELINE ESTIMATES

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.252*** (0.096)	0.235*** (0.090)	0.244*** (0.071)	0.225*** (0.068)
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Bond controls	No	Yes	No	Yes
City pre-shock var. $\times$ Post	No	No	Yes	Yes
Adjusted R^2	0.506	0.568	0.516	0.578
Obs	87885	87885	87885	87885

Notes: This table reports the regression results of the impact of GSA on MCB spreads based on the specification of Equation (1). The dependent variable is calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year-quarter and issuer fixed effects. Bond controls contain (ln)bond issuing amount, bond maturity, bond rating, issuer rating, and whether the bond is option-embedded, guaranteed, and traded on exchange. City pre-shock variables include (ln)GDP, (ln)GDP per capita, GDP growth rate, the proportion of tertiary sector in GDP, (ln)population, population growth rate, (ln)nighttime light intensity, (ln)fixed investment, and (ln)housing price at city level in 2013. Detailed definitions of control variables are presented in Table A1. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 4: GSA AND MCB SPREADS: PLACEBO TEST

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.225*** (0.069)	0.223*** (0.070)	0.221*** (0.070)	0.218*** (0.072)
ProvNR $\times$ Post	-0.033 (0.088)			-0.034 (0.087)
CityNR $\times$ Post		-0.013 (0.072)		-0.017 (0.076)
CountyNR $\times$ Post			0.024 (0.074)	0.030 (0.077)
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.578	0.578	0.578	0.578
Obs	87885	87885	87885	87885

Notes: This table reports the impact of owning different types of nature reserves on MCB spreads around GSA. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one in and after 2017 and zero otherwise. *ProvNR* is a dummy variable that equals one if there is at least one provincial nature reserve in the city where the issuer is located and zero otherwise. *CityNR* is a dummy variable that equals one if there is at least one municipal nature reserve in the city where the issuer is located and zero otherwise. *CountyNR* is a dummy variable that equals one if there is at least one county nature reserve in the city where the issuer is located and zero otherwise. Regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 5: GSA AND LOCAL FISCAL CONDITION

Dep. Var	$\frac{\text{Deficit}_t}{\text{Revenue}_{2013}}$	$\frac{\text{Deficit}_t}{\text{GDP}_{2013}}$	$\frac{\text{Expense}_t}{\text{GDP}_{2013}}$	$\frac{\text{Revenue}_t}{\text{GDP}_{2013}}$
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.188** (0.087)	1.679*** (0.558)	1.399** (0.596)	-0.449** (0.222)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.960	0.962	0.960	0.886
Obs	2725	2725	2725	2725

Notes: This table reports the impact of GSA on local fiscal conditions at the city-year level. The dependent variable is the fiscal deficit (annual fiscal expenditures minus revenues), divided by annual fiscal revenue in 2013 in Column (1), fiscal deficit divided by city GDP in 2013 (multiplied by 100) in Column (2), annual fiscal expenditure divided by city GDP in 2013 (multiplied by 100) in Column (3), and annual fiscal revenue divided by city GDP (multiplied by 100) in 2013 in Column (4), respectively. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year and city fixed effects, as well as the city-level control variables (i.e., *city pre-shock var. $\times$ Post* used in Table 3). Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 6: GSA AND PUBLIC PROCUREMENT RELATED TO TRANSITION EFFORTS

Dep. Var	Narrow definition		Broad definition	
	Number _t	Value _t	Number _t	Value _t
	GDP ₂₀₁₃	GDP ₂₀₁₃	GDP ₂₀₁₃	GDP ₂₀₁₃
	(1)	(2)	(3)	(4)
NNR × Post	0.014*** (0.002)	0.004*** (0.001)	0.077*** (0.018)	0.023*** (0.007)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.595	0.435	0.659	0.511
Obs	2725	2725	2725	2725

Notes: This table reports the impact of GSA on local public procurement related to biodiversity transition efforts at the city-year level. Government procurement deals are identified according to the narrow definition in Columns (1)-(2) and broad definition in Columns (3)-(4), respectively. The dependent variable is the number of procurement deals divided by city GDP in 2013 (the unit of GDP is in billions of CNY) in Columns (1) and (3), and the monetary value of procurement deals divided by city GDP in 2013 (the ratio is multiplied by 100) in Columns (2) and (4), respectively. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year and city fixed effects, as well as the city-level control variables (i.e., *city pre-shock var.* × *Post* used in Table 3). Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 7: GSA AND MCB SPREADS: PRE-EXISTING ECONOMIC ACTIVITIES IN NNRs

Dep. Var	Spread	
	Developed land	Nighttime light
	(1)	(2)
Grouping reference		
NNR $\times$ Post $\times$ High ex-ante econ activity in NNR	0.265** (0.123)	0.195* (0.111)
NNR $\times$ Post	0.173** (0.072)	0.177** (0.073)
Other terms of triple differences	Yes	Yes
Controls	Yes	Yes
Year-quarter FE	Yes	Yes
Issuer FE	Yes	Yes
Adjusted R^2	0.578	0.578
Obs	87885	87885

Notes: This table reports the estimated heterogeneous effects of GSA on MCB spreads in pre-existing human economic activities within NNRs. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *High ex-ante economic activity in NNRs* is a dummy variable that equals one if a city is in the top quartile of the intensity distribution of human economic activities within NNRs in 2016 among NNR municipalities. The intensity of human economic activities is measured by the developed land area within NNRs in Column (1) and nighttime light intensity within NNRs in Column (2), respectively. Regressions include all required components of the triple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 8: GSA AND MCB SPREADS: HETEROGENEOUS EFFECTS BY INFORMATION AVAILABILITY PRE-GSA

Dep. Var	Spread				
	Full sample	Drop single-news city	NNR news cover=1	NNR news cover=0	Full sample
	(1)	(2)	(3)	(4)	(5)
NNR city sample selection					
NNR $\times$ Post	0.231** (0.103)	0.228** (0.101)	0.136* (0.075)	0.190* (0.111)	0.189* (0.112)
NNR $\times$ Post $\times$ NNR News cover	-0.008 (0.102)	0.015 (0.102)			-0.023 (0.113)
NNR $\times$ Post $\times$ High ex-ante econ act in NNR			0.253* (0.145)	0.320* (0.184)	0.340** (0.167)
NNR $\times$ Post $\times$ High ex-ante econ act in NNR $\times$ NNR News cover					-0.089 (0.217)
Other terms of triple/quadruple differences	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.578	0.577	0.581	0.569	0.578
Obs	87885	75184	74572	56253	87885

Notes: This table reports the heterogeneous effects of GSA on MCB spreads in information asymmetry. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *NNR News cover* is a dummy variable that equals one if NNRs managed by an NNR municipality were covered by the top 10 newspapers by average circulation in 2016. Column (1) employs the full sample. Column (2) excludes cities that owned NNR and only covered by one source of the top 10 newspapers by average circulation in 2016, for robustness. Columns (3) and (4) replicate Column (1) of Table 7, restricting the NNR cities to those with and without NNR-related media coverage, respectively, while keeping the non-NNR cities unchanged and fully included in the regressions. Column (5) reports a quadruple-difference specification estimated on the full sample. Regressions include all required components of the triple/quadruple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE 9: GSA AND BIRD-WATCHING OBSERVATIONS

Dep. Var	# of bird species observed	# of birdwatching reporters	# of birdwatching reports
	(1)	(2)	(3)
NNR $\times$ Post	8.987* (4.568)	0.771 (0.727)	2.403 (4.573)
Year-quarter FE	Yes	Yes	Yes
City FE	Yes	Yes	Yes
Adjusted R^2	0.606	0.576	0.511
Obs	4582	4582	4582

Notes: This table reports the city-quarter-level regression results of the effects of GSA on the birdwatching observations reported on the China Bird Report Center. The dependent variables in Columns (1)-(3) are the number of bird species observed, the number of birdwatching reporters and the number of birdwatching reports, respectively. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year-quarter and city fixed effects. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

INTERNET APPENDIX

A ADDITIONAL TABLES AND FIGURES

FIGURE A1: CHANGES IN THE ECOLOGICAL CONDITION OF THE QILIAN MOUNTAIN NATIONAL NATURE RESERVE AROUND GSA

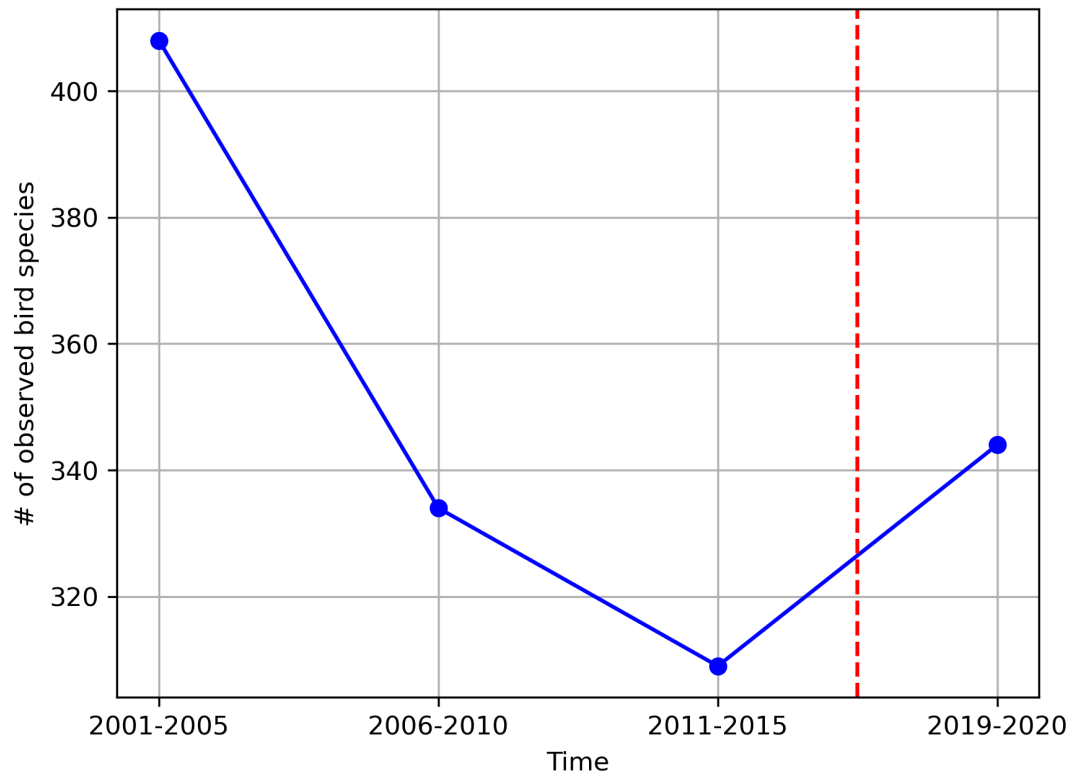


Pre-GSA period

Post-GSA period

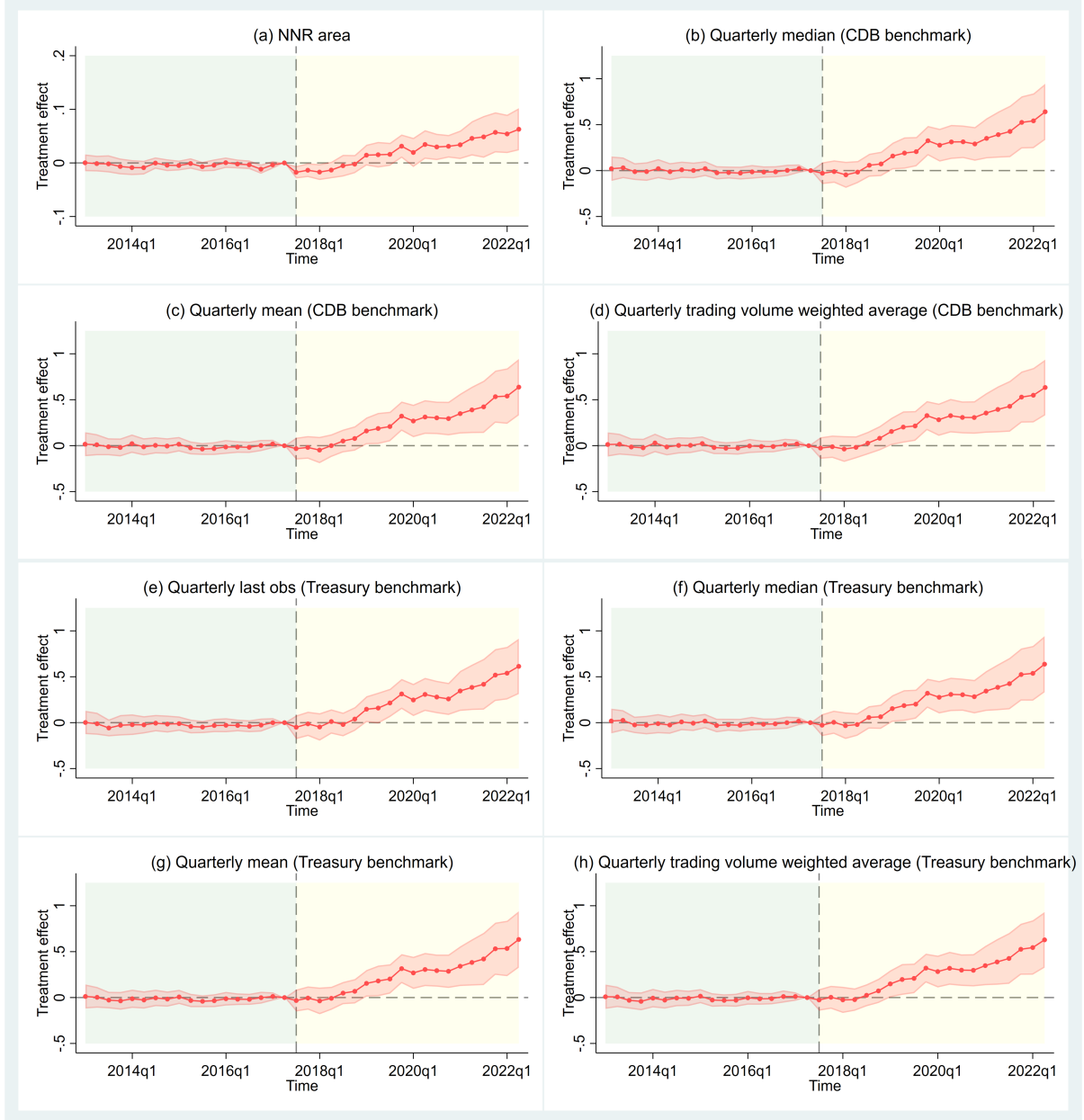
Notes: This figure illustrates the contrast in the ecological condition within the Qilian Mountain National Nature Reserve before and after GSA, from the perspectives of the disposal of illegal hydropower facilities (the upper part) and colliery (the bottom part). See more details in: https://www.mee.gov.cn/ywgz/zysthjbjhdc/dczg/202102/t20210206_820575.shtml.

FIGURE A2: THE DYNAMICS OF BIRD SPECIES OBSERVED OVER TIME



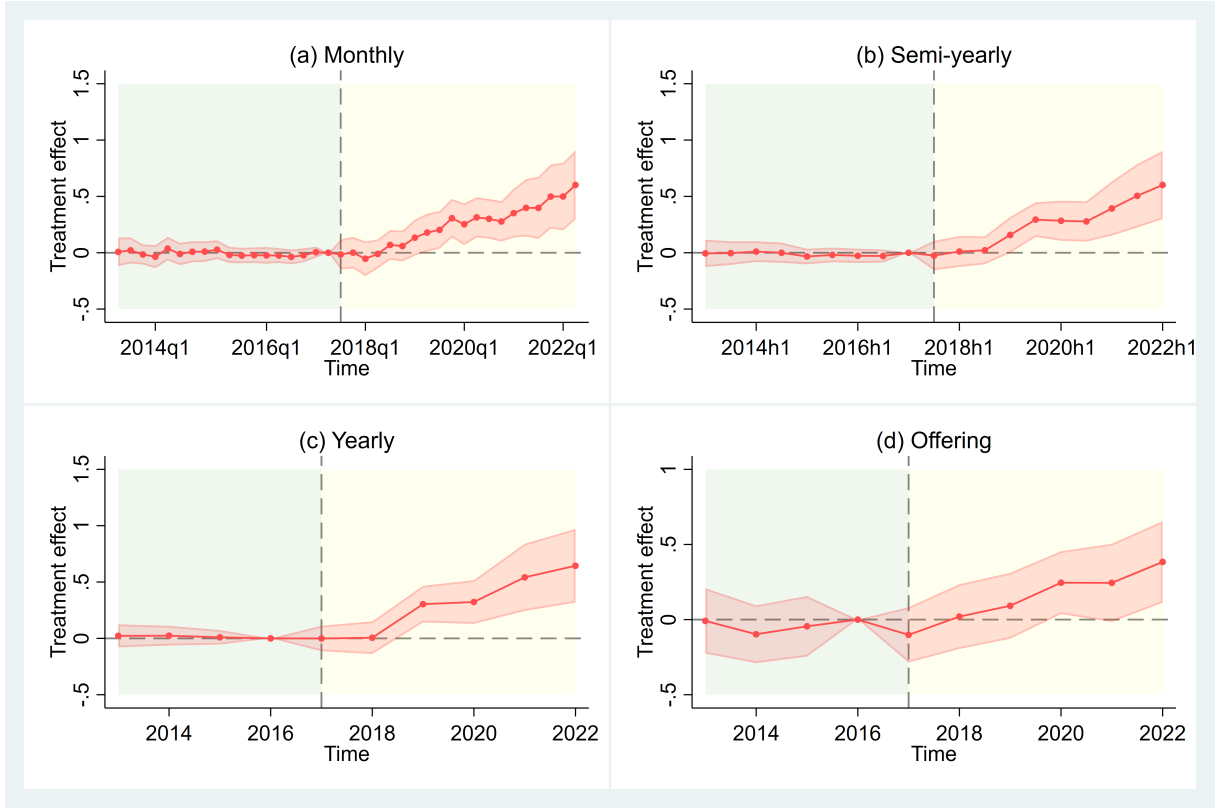
Notes: This figure depicts the total number of bird species surveyed every five years since the beginning of this century by the forest ecological stations connected with the Chinese National Ecosystem Research Network (CNERN). The dash vertical red line represents the time when GSA was launched.

FIGURE A3: CHANGES IN MCB SPREADS BEFORE AND AFTER GSA:
ALTERNATIVE MEASURES



Notes: This figure depicts results of robustness tests on estimated differences in MCB spreads between NNR municipalities and non-NNR municipalities, by replacing the measures of main variables in Figure 3. Panel (a) presents the result using an alternative measure of treatment intensity: the proportion of NNRs in the area of urban administrative areas within a city. Panel (b)-(h) present the results using alternative measures of spreads: Panel (b)-(d) employ the quarterly median, the quarterly mean, and the quarterly trading-volume-weighted average of MCB spreads, respectively, using the risk-free benchmark of CDB yield; Panel (e)-(h) replace the risk-free benchmark with the Treasury yield.

FIGURE A4: CHANGES IN MCB SPREADS BEFORE AND AFTER GSA:
ALTERNATIVE FREQUENCIES OF BOND SAMPLE



Notes: This figure depicts the results of robustness tests on estimated differences in MCB spreads between NNR municipalities and non-NNR municipalities before and after GSA, by replacing the data frequency of [Figure 3](#). The alternative frequencies of the bond sample are constructed by choosing the last observation of monthly frequency, semi-yearly frequency, and yearly frequency for each bond in Panel (a)-(c), respectively. The estimation result with the bond offering sample is also shown in Panel (d).

TABLE A1: DEFINITIONS OF VARIABLES

Variable	Definition
Bond size	The bond issuing amount, in hundred million yuan RMB.
Time to maturity	The residual maturity of a bond traded, in year.
Option	A dummy variable that equals one if a bond is option-embedded and zero otherwise.
Guarantee	A dummy variable that equals one if a bond is guaranteed and zero otherwise.
Exchange	A dummy variable that equals one if a bond is traded on the exchange market and zero otherwise.
Bond rating	A numerical number converted from the real letter grades of a bond, by assigning 1 to AAA, 2 to AA+, 3 to AA, 4 to AA-, 5 to A+, 6 to A, 7 to A-, 8 to grades below A.
Issuer rating	A numerical number converted from the real letter grades of an issuer, by assigning 1 to AAA, 2 to AA+, 3 to AA, 4 to AA-, 5 to A+, 6 to A, 7 to A-, 8 to grades below A.
GDP	The gross domestic product of the corresponding city in 2013, in ten billion yuan RMB.
GDP per capita	The GDP per capita of the corresponding city in 2013, in thousand yuan RMB.
GDP annual growth rate	The ratio of the difference between GDP in 2013 and that in 2012 to the GDP in 2012 of the corresponding city, in percentage.
Tertiary sector GDP	The ratio of the tertiary sector in GDP of the corresponding city in 2013, in percentage.
Nighttime lights	The nighttime light intensity of the corresponding city in 2013.
Fixed investments	The fixed investments of the corresponding city in 2013, in ten billion RMB yuan.
Housing price	The average selling price of commercial housing of the corresponding city in 2013, in thousand RMB yuan per square meter.
Population	The permanent resident population of the corresponding city in 2013, in million people.
Population annual growth rate	The ratio of the difference between permanent resident population in 2013 and that in 2012 to the permanent resident population in 2012 of the corresponding city, in percentage.
Urbanization rate	The ratio of non-agricultural residents to the total population of the corresponding city in 2010, in percentage.
High school and above education rate	The ratio of population with high school education or above to the total population of the corresponding city in 2010, in percentage.
Local fiscal revenue	The general budget revenue of the corresponding city government in 2013, in million yuan RMB.
Local fiscal expenditure	The general budget expenditure of the corresponding city government in 2013, in million yuan RMB.

Notes: This table reports the detailed definitions of baseline controls (see Table 3) and balance-test variables (see Table 2). Bond-level variables are collected from WIND. The intensity of nighttime lights at the city level are calculated based on the raster data of Zhang *et al.* (2024). Urbanization and high school education rates are collected from the 2010 census (In China, censuses are conducted every ten years and many city governments do not release detailed demographic data annually). Other city-level variables are collected by the National Bureau of Statistics and local statistical yearbooks in China. The exchange rate between the US dollar and the CNY was typically between 1:6 and 1:7.2.

TABLE A2: GSA AND PUBLIC FINANCING DEMAND

Dep. Var	MCB issuance dummy	MCB issuance amount	Growth rate of LGFV debt	Growth rate of LGFV interest-bearing debt
	(1)	(2)	(3)	(4)
NNR $\times$ Post	-0.044 (0.030)	-1.702 (1.285)	-1.107 (3.744)	-8.326 (6.273)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.372	0.775	0.180	0.154
Obs	2437	2437	2437	2437

Notes: This table reports the city-year-level regression results of the effects of GSA on the quantity of local public financing. *MCB issuance dummy* is a dummy variable that equals one if a city has a new MCB issuance in that year and zero otherwise. *MCB issuance amount* is a continuous variable that represents the total amount of new MCBs issued by a city in that year (in billion CNY). *Growth rate of LGFV debt* is a continuous variable that represents the growth rate of city-year-level aggregated total debts of LGFVs with outstanding MCBs (in percentage). *Growth rate of LGFV interest-bearing debt* is a continuous variable that represents the growth rate of city-year-level aggregated interest-bearing debts of LGFVs with outstanding MCBs (in percentage). *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city and zero otherwise. *Post* is a dummy variable that equals one for years in and after 2017 and zero otherwise. Regressions include year and city fixed effects, as well as the city-level control variables (i.e., *city pre-shock var.* $\times$ *Post* used in Table 3). Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A3: GSA AND LOCAL POLITICAL RISK

Panel A: GSA and local political turnover		
Dep. Var	Turnover dummy	
	Municipal CPC secretary	Mayor
	(1)	(2)
NNR $\times$ Post	-0.026 (0.032)	0.044 (0.028)
Controls	Yes	Yes
Year FE	Yes	Yes
City FE	Yes	Yes
Adjusted R^2	0.093	0.116
Obs	2725	2725

Panel B: GSA and MCB spreads: political incentive heterogeneity		
Dep. Var	Spread	
	Municipal CPC secretary	Mayor
	(1)	(2)
NNR $\times$ Post	0.228*** (0.076)	0.225*** (0.075)
NNR $\times$ Post $\times$ In the first 2 years of tenure	-0.002 (0.056)	0.007 (0.059)
Other terms of triple differences	Yes	Yes
Controls	Yes	Yes
Year-quarter FE	Yes	Yes
Issuer FE	Yes	Yes
Adjusted R^2	0.578	0.578
Obs	87885	87885

Notes: This table reports the relationship between GSA and the local political environment. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city and zero otherwise in both panels. Panel A reports the effects of GSA on local political turnover at the city-year level. The dependent variable in Panel A is a dummy variable that equals one if the corresponding city's leading official changes in that year. *Post* in Panel A is a dummy variable that equals one in and after 2017 and zero otherwise. Regressions in Panel A include year and city fixed effects, as well as the city-level control variables (i.e., *city pre-shock var. $\times$ Post* used in Table 3). Panel B reports the differential impacts of GSA during specific stages of the official's tenure. The dependent variable in Panel B is MCB spread. *Post* in Panel B is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *In the first 2 years of tenure* is a dummy variable that equals one if it is in the first two complete years of the corresponding official's term and zero otherwise. Regressions in Panel B include all required components of the triple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses of both panels are all clustered at the city level. ⁶***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A4: GSA AND BIODIVERSITY IMPROVEMENT: IUCN MEASURE

Dep. Var	species richness - mean	species richness - rank
	(1)	(2)
NNR $\times$ Post	11.781*** (3.692)	-10.851*** (3.429)
Year FE	Yes	Yes
City FE	Yes	Yes
Adjusted R^2	0.962	0.950
Obs	624	624

Notes: This table reports the city-year-level regression results of the effects of GSA on local biodiversity using IUCN Red List data. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city and zero otherwise. *Post* is a dummy variable that equals one in 2022 and zero in 2017. The dependent variable is the average number of species potentially occurring per grid in the city in Column (1), and the city's species richness ranking according to this average in the corresponding year in Column (2). Regressions include year and city fixed effects. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE A5: GSA AND MCB SPREADS: BIODIVERSITY IMPROVEMENT

Dep. Var	Spread			
	Birds species observed		Species - ICUN measure	
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.178** (0.080)	0.189*** (0.071)	0.197** (0.080)	0.216*** (0.080)
NNR $\times$ Post $\times$ High bio improvement	0.115 (0.105)	0.169 (0.172)	0.116 (0.107)	0.035 (0.094)
Other terms of triple differences	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.576	0.576	0.578	0.578
Obs	86039	86039	87885	87885

Notes: This table reports the heterogeneous effects of GSA on MCB spreads in biodiversity improvement. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *High bio improvement* is a dummy variable that equals one if a city is in the top quartile of the biodiversity improvement among NNR municipalities. In Column (1) and (3)-(4), the level of biodiversity improvement is measured by the absolute increase in the number of species following GSA for each NNR municipality (i.e., the number of species after GSA - that before GSA); In Column (2), the level of biodiversity improvement is measured by the relative increase in number of species following GSA for each NNR municipality (i.e., (the number of species after GSA - that before GSA) / that before GSA). Consistent with the description in [Section 6.1](#), the number of species is calculated from the city's value of bird species observed based on birdwatching data in Columns (1)-(2), the city's value of average species richness based on IUCN Red List data in Column (3), and the city's rank of average species richness based on IUCN Red List data in Column (4). Regressions include all required components of the triple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in [Table 3](#). Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A6: GSA AND MCB SPREADS: MUTUAL FUND HOLDING

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post $\times$ Top bond holdings by mutual funds	-0.046 (0.056)	-0.053 (0.056)	-0.050 (0.055)	-0.054 (0.057)
NNR $\times$ Post	0.230*** (0.071)	0.230*** (0.071)	0.232*** (0.071)	0.231*** (0.070)
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.579	0.599	0.604	0.600
Obs	87885	87885	87885	87259

Notes: This table reports the heterogeneous effects of GSA on MCB spreads in mutual fund holding. Specifically, we use the spread on the last trading day of each quarter in Column (1) (the same as that in Table 3), and the measures used are the quarterly median of spreads, the quarterly mean of spreads, and the quarterly trading-volume-weighted spreads in Columns (2)-(4), respectively (as explained in Section 7). *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *Top bond holdings by mutual funds* is a dummy variable that equals one if the bond was disclosed as a top holding by a mutual fund in the quarter preceding the trade. Regressions include all required components of the triple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A7: GSA AND MCB SPREADS: WEIGHTED LEAST SQUARE REGRESSION
ACCORDING TO PRE-GSA CITY DEBT LEVEL

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.327** (0.137)	0.378*** (0.120)	0.328*** (0.094)	0.358*** (0.087)
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Bond controls	No	Yes	No	Yes
City pre-shock var. $\times$ Post	No	No	Yes	Yes
Adjusted R^2	0.430	0.517	0.440	0.526
Obs	87643	87643	87643	87643

Notes: This table reports the regression results of the impact of GSA on MCB spreads using weighted least square estimation. The weighting factor is the city-level average value of LGFV interest-bearing debts from 2013 to 2016. The dependent variable is calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year-quarter and issuer fixed effects. Bond controls contain (ln)bond issuing amount, bond maturity, bond rating, issuer rating, and whether the bond is option-embedded, guaranteed, and traded on exchange. City pre-shock variables include (ln)GDP, (ln)GDP per capita, GDP growth rate, the proportion of tertiary sector in GDP, (ln)population, population growth rate, (ln)nighttime light intensity, (ln)fixed investment, and (ln)housing price at city level in 2013. Detailed definitions of control variables are presented in [Table A1](#). Standard errors in parentheses are clustered at city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A8: ADDITIONAL LOCAL PUBLIC DEBT COSTS AND CORRESPONDING ECONOMIC SIGNIFICANCE

Scenario A: China bears the gap in biodiversity financing according to its share of global species (6.4%)				
Scenario B: China bears the gap in biodiversity financing according to its share of global land area (7%)				
Scenario C: China bears the gap in biodiversity financing according to its share of global GDP (18.5%)				
Year	2018	2019	2020	2021
Additional financing costs for MBC markets (Billion dollars)	1.03	5.54	11.51	21.37
Global biodiversity financing gap estimated by Deutz <i>et al.</i> (2020) (Billion dollars)		711	711	711
China's biodiversity financing gap in Scenario A (Billion dollars)		45.5	45.5	45.5
China's biodiversity financing gap in Scenario B (Billion dollars)		49.77	49.77	49.77
China's biodiversity financing gap in Scenario C (Billion dollars)		131.54	131.54	131.54
The proportion of additional financing costs in China's biodiversity financial gap in Scenario A (%)		12.18	25.30	46.97
The proportion of additional financing costs in China's biodiversity financial gap in Scenario B (%)		11.13	23.13	42.94
The proportion of additional financing costs in China's biodiversity financial gap in Scenario C (%)		4.21	8.75	16.24

Notes: This table presents results of a simple back-of-the-envelope calculation on the aggregate costs of GSA on the LGFVs' debt interest payments and the corresponding economic significance. The exchange rate between the US dollar and the CNY is set as 1: 7.

TABLE A9: GSA AND MCB SPREADS: ALTERNATIVE SAMPLES

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.208*** (0.073)	0.243*** (0.068)	0.274*** (0.075)	0.246*** (0.079)
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.578	0.581	0.580	0.580
Obs	80978	84623	76890	69435

Notes: This table reports the regression results of the impact of GSA on MCB spreads using alternative samples of NNR and non-NNR municipalities. Column (1) excludes NNR municipalities in the top quartile of the distribution of *NNR area* described in [Section 7](#) in 2016. Column (2) uses only non-NNR municipalities adjacent to an NNR municipality as control group. Column (3) uses only non-NNR municipalities within a 50 km radius from an NNR municipality as control group. Column (4) uses the intersection set of samples from Columns (1)-(3). The dependent variable is the MCB spread, calculated as the difference between the MCB yield and the CDB yield on the same day and of the same maturity. The variable *NNR* is a dummy that equals one if there is at least one national nature reserve in the city where the issuer is located, and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017, and zero otherwise. Regressions include year-quarter and issuer fixed effects, as well as all controls defined in [Table 3](#). Detailed variable definitions are presented in [Table A1](#). Standard errors clustered at the city level are shown in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE A10: GSA AND MCB SPREADS: CENTRAL INSPECTION ON ENVIRONMENTAL PROTECTION

Dep. Var	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
NNR $\times$ Post	0.225*** (0.068)	0.225*** (0.068)	0.225*** (0.068)	0.222*** (0.069)	0.218*** (0.069)	0.215*** (0.069)
In the 1st round	0.054** (0.024)		0.054** (0.024)			
In the 2nd round		-0.033 (0.042)	-0.033 (0.042)			
After the 1st round				0.082** (0.036)		0.079** (0.036)
After the 2nd round					0.081 (0.062)	0.079 (0.062)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.578	0.578	0.578	0.578	0.578	0.578
Obs	87885	87885	87885	87885	87885	87885

Notes: This table reports the effects of the Central Inspection on Environmental Protection (CIEP) on MCB spreads. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *In the 1st (2nd) round* is a dummy variable that equals one if the province is under the first-round (second-round) investigation of CIEP and zero otherwise. *After the 1st (2nd) round* is a dummy variable that equals one for provinces investigated in the first (second) round after the first quarter of investigation and zero otherwise. All regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A11: GSA AND MCB SPREADS: NATIONWIDE BATTLE TO PREVENT AND CONTROL POLLUTION

Dep. Var	Spread					
	(1)	(2)	(3)	(4)	(5)	(6)
NNR $\times$ Post	0.195*** (0.072)	0.220*** (0.070)	0.212*** (0.071)	0.211*** (0.071)	0.204*** (0.073)	0.179** (0.076)
AQI $\times$ Post2018Q1	-0.003 (0.003)					-0.004 (0.003)
Industrial SO2 / GDP2 $\times$ Post2018Q1		0.978*** (0.179)			1.054*** (0.257)	1.040*** (0.245)
industrial sewage / GDP2 $\times$ Post2018Q1			-0.054 (0.164)		-0.275* (0.165)	-0.294* (0.164)
Industrial dust / GDP2 $\times$ Post2018Q1				0.625** (0.243)	0.087 (0.295)	0.104 (0.287)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.568	0.565	0.561	0.561	0.563	0.563
Obs	85741	84739	84628	83482	83482	83482

Notes: This table reports the effects of the Nationwide Battle to Prevent and Control Pollution (NBPCP) on MCB spreads. The dependent variable is MCB spread, calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *AQI* is the Annual Air Quality Index at city level in 2017. *Industrial SO2* represents annual industrial sulfur dioxide emissions at city level in 2017. *Industrial sewage* represents annual industrial wastewater emissions at city level in 2017. *Industrial dust* represents annual industrial dust emissions at city level in 2017. *GDP2* represents gross domestic product of the secondary sector at city level in 2017. *PostNBPCP* is a dummy variable that equals one after the first quarter of 2018 and zero otherwise. All regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A12: GSA AND MCB SPREADS: AAA-RATED BONDS

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.178** (0.081)	0.152** (0.076)	0.149* (0.076)	0.156** (0.076)
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.674	0.687	0.698	0.695
Obs	17920	17920	17920	17676

Notes: This table reports the regression results of the impact of GSA on MCB spreads, excluding bonds with credit ratings below AAA in Table 3. The dependent variable is calculated as the difference between the MCB yield and CDB yield on the same day and of the same maturity. Specifically, we use the spread on the last trading day of each quarter in Column (1) (the same as that in Table 3), and the measures used are the quarterly median of spreads, the quarterly mean of spreads, and the quarterly trading-volume-weighted spreads in Columns (2)-(4), respectively (as explained in Section 7). *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A13: GSA AND CSOE BOND SPREAD

Dep. Var	Spread			
	(1)	(2)	(3)	(4)
NNR $\times$ Post	0.027 (0.093)	0.021 (0.094)	0.030 (0.095)	0.026 (0.094)
Controls	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.734	0.748	0.742	0.744
Obs	23824	23824	23824	23824

Notes: This table reports the regression results of the impact of GSA on CSOE bond spreads. The dependent variable is calculated as the difference between the CSOE bond yield and CDB yield on the same day and of the same maturity. Specifically, we use the spread on the last trading day of each quarter in Column (1) (the same as that in Table 3), and the measures used are the quarterly median of spreads, the quarterly mean of spreads, and the quarterly trading-volume-weighted spreads in Columns (2)-(4), respectively (as explained in Section 7). *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A14: GSA AND MCB SPREADS: LOCAL PUBLIC DEBT BURDEN

Dep. Var Grouping reference	Spread	
	Total debt	Interest-bearing debt
	(1)	(2)
NNR $\times$ Post $\times$ High public debt burden	0.244** (0.119)	0.291** (0.127)
NNR $\times$ Post	0.109 (0.088)	0.083 (0.094)
Other terms of triple differences	Yes	Yes
Controls	Yes	Yes
Year-quarter FE	Yes	Yes
Issuer FE	Yes	Yes
Adjusted R^2	0.578	0.580
Obs	87837	87837

Notes: This table reports the heterogeneous effects of GSA on MCB spreads in local public debt burden. The dependent variable is MCB spread. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city where the issuer is located and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. *High debt burden* is a dummy variable that equals one if a city's level of public debt burden is in the top quartile of the city distribution for the year before the bond trade and zero otherwise, where the debt burden is measured by city-year-level total debt in Columns (1) and interest-bearing debt in Columns (2) of LGFVs, divided by city GDP, respectively. Regressions include all required components of the triple-difference model. All regressions all required components of the triple-difference model, year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses of both panels are all clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

TABLE A15: GSA AND MCB SPREADS: BOND MATURITY

Dep. Var	Spread					
Grouping indicator	Long Term >3 years			Long Term >4 years		
Sample	Short Term	Long Term	Full Sample	Short Term	Long Term	Full Sample
	(1)	(2)	(3)	(4)	(5)	(6)
NNR $\times$ Post	0.318*** (0.094)	0.178*** (0.063)	0.302*** (0.086)	0.290*** (0.073)	0.155** (0.075)	0.271*** (0.074)
NNR $\times$ Post $\times$ Long Term			-0.158** (0.070)			-0.154** (0.063)
Other terms of triple differences	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.595	0.672	0.579	0.589	0.688	0.578
Obs	35260	52625	87885	48414	39471	87885

Notes: This table reports the estimated heterogeneous effects of GSA on MCB spreads in bond maturity. The criterion to classify between long-term and short-term bond is whether is residual maturity is above 3 years in Columns (1)-(3) and 4 years in Columns (4)-(6), respectively. *NNR* is a dummy variable that equals one if there is at least one national nature reserve in the city and zero otherwise. *Post* is a dummy variable that equals one after the second quarter of 2017 and zero otherwise. Column (3) and (6) include all required components of the triple-difference model. All regressions include year-quarter and issuer fixed effects, as well as all controls defined in Table 3. Standard errors in parentheses are clustered at the city level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

B ADDITIONAL MATERIALS ON INSTITUTIONAL BACKGROUND AND DATA

B.1 Two Examples of Government Procurement on NNRs

FIGURE B1: GOVERNMENT PROCUREMENT RELATED TO NNR (1)

甘肃尕斯海则岔国家级自然保护区管理局2020年尕斯海保护区草原生态修复治理（退化草原植被修复）项目成交公告		Name of the NNR																																																														
2020-08-19 来源：中国政府采购网 【查看原文】 【显示公告概要】																																																																
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Notes: This procurement project aims to restore the grassland ecosystem in the Gansu Gahai-Zacha NNR by rehabilitating degraded vegetation. The announcement was made on August 19, 2020. The procuring government department is the Gansu Gahai-Zacha National Nature Reserve Administration. The amount of this project is 3,370,680 CNY.

FIGURE B2: GOVERNMENT PROCUREMENT RELATED TO NNR (2)

广东内伶仃福田国家级自然保护区管理局基于环志和卫星追踪技术对福田红树林鸟类栖息地生态修复的影响评估成交公告

2021-12-08 来源: 中国政府采购网 【查看原文】 【显示公告概要】

公告概要:

公告信息:	
采购项目名称	基于环志和卫星追踪技术对福田红树林鸟类栖息地生态修复的影响评估
品目	服务/其他服务
采购单位	广东内伶仃福田国家级自然保护区管理局
行政区域	深圳市
公告时间	2021年12月08日 18:52
评审专家 (单一来源采购人员) 名单	刘多、袁晖、朱起德
总成交金额	¥79.500000 万元 (人民币)
联系人及联系方式:	
项目联系人	黄工、王工
项目联系电话	0755-82077536转107、103
采购单位	广东内伶仃福田国家级自然保护区管理局
采购单位地址	深圳市福田区车公庙红树林路1号
采购单位联系方式	杨工 0755-83225610, 83708803
代理机构名称	中国远东国际招标有限公司
代理机构地址	深圳市福田区上步南路1001号锦峰大厦22楼
代理机构联系方式	杜工 0755-82078819、82077364转101

Name of the NNR

Name of the procurement project

Name of the government department

Announcement time

Amount of the project

Notes: This government procurement deal is about a project assessing the effectiveness of ecological restoration efforts for bird habitats in the Futian Mangrove, employing bird banding and satellite tracking technologies. The announcement was made on December 8, 2021. The procuring government department is the Guangdong Neilingding Futian National Nature Reserve Administration. The amount of this project is 795,000 CNY.

B.2 Biodiversity-related Keyword Dictionary

TABLE B1: BIODIVERSITY-RELATED KEYWORD DICTIONARY

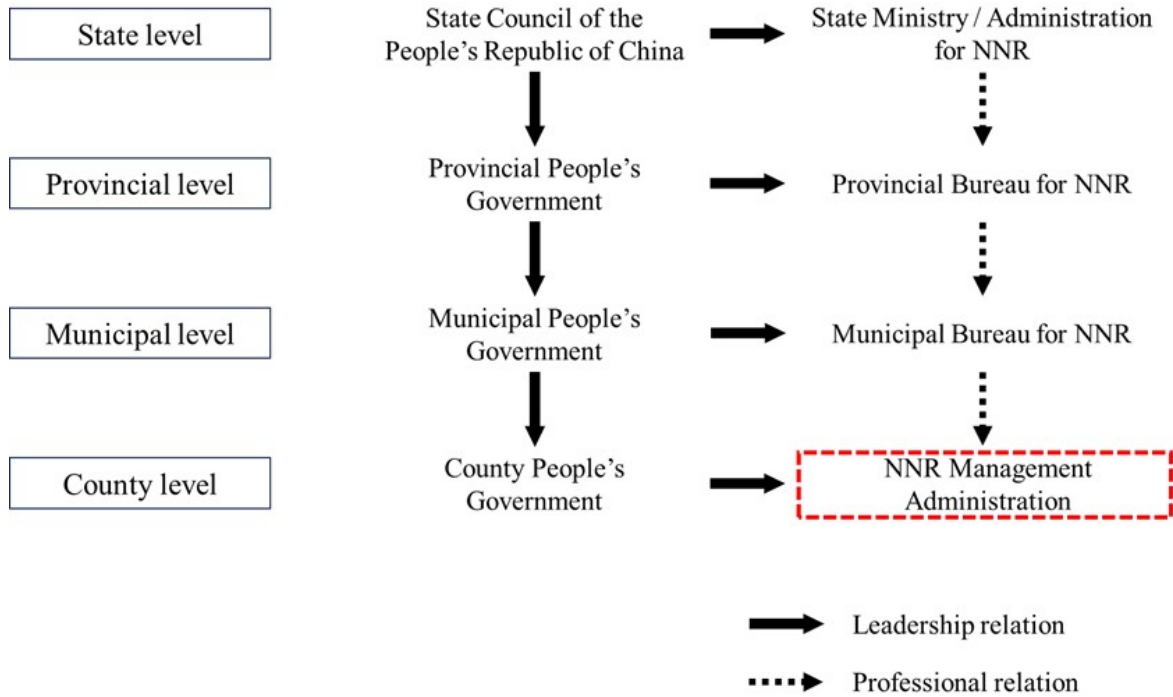
Source Chinese Keywords	Translated English Keywords
保护地	Protected Area
自然	Nature
物种	Species
鸟类	Bird
野生	Wild
天然林	Natural Forest
渔业资源	Fishery Resource
海岛	Island
生态系统	Ecosystem
海域	Sea Area
涵养林	Conservation Forest
海洋	Ocean
保护区	Nature Reserve
公益林	Public Welfare Forest
雨林	Rainforest
自然生态	Natural Ecology
湿地	Wetland
栖息地	Habitat
森林	Forest
黑土地	Black Soil
水域	Water Area
野外	Wilderness
湖泊	Lake
动植物	Flora and Fauna
珍稀	Rare
生物多样性	Biodiversity
濒危	Endangered
鱼类	Fish
红树林	Mangrove
生境	Biotope

Notes: This table presents the keyword dictionary for identifying broad-scope biodiversity procurement.

B.3 Institutional Arrangements for NNR Management Administration

In China, local governments are primarily responsible for managing NNRs within their jurisdictions. Each NNR is typically overseen by a dedicated management administration that handles its routine operations. This governance structure operates under a “dual leadership” model, which encompasses two distinct types of leadership: the “leadership relationship,” where the local territorial government controls the personnel and financial budget of the NNR administration, and the “professional relationship,” where higher-level government agencies responsible for NNR management (e.g., the forestry bureau) supervise the specialized work content of lower-level agencies to ensure they fulfill their designated functions effectively. Consequently, NNR management is primarily accountable to local governments rather than higher-level authorities (Wang *et al.*, 2023). The institutional structure of NNR organizations is illustrated in Figure B3 below.

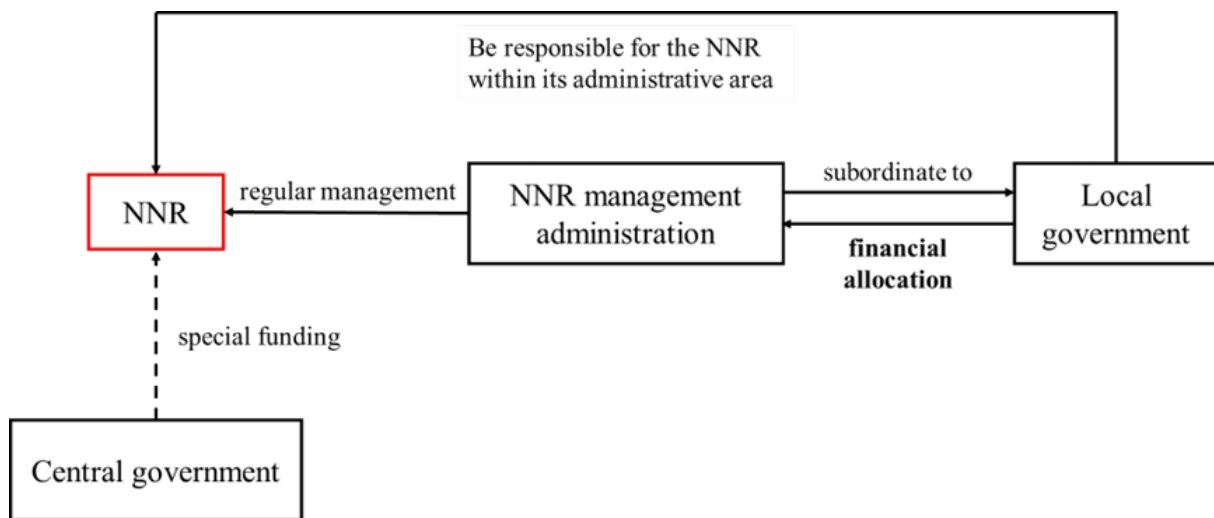
FIGURE B3: INSTITUTIONAL ARRANGEMENTS FOR NNR ADMINISTRATION



B.4 Funding Sources for NNRs

Consistent with the [Section 2](#) and [Figure B3](#), the operational oversight and financial sustenance of NNRs predominantly fall under the purview of local governments in China. In contrast, the allocation of resources from the central government is relatively limited. [Figure B4](#) presents the institutional arrangement pertaining to the funding of NNRs:

FIGURE B4: THE STRUCTURE OF FUNDING FOR NNR



Here are additional materials that can support that local governments provide the majority of funds to NNRs:

1. Data from the National Forestry and Grassland Administration of PRC

According to China Forestry and Grassland Statistical Yearbook, the total investment in the “Nature Reserve Monitoring and Management” project in 2019 was 1.839 billion CNY, of which the funding from the central government was 0.704 billion CNY, accounting for 38.28% of the total investment. In the subsequent years of 2020, 2021, and 2022, the proportions of central government funding were 46.22%, 38.59%, and 41.88% respectively.

2. Some fragments of a report from “YICAI” magazine in 2019

Based on the interview of the 13th National People’s Congress deputy Jianbo Sun by “YICAI” magazine in 2019, the central government’s funding allocation for NNRs in China is very limited.

“Nature reserves are common goods of the country. Ensuring funding is an inevitable requirement for the government to fulfill its responsibility for managing nature reserves,”

said Jianbo Sun. However, in reality, the central government only provides a small amount of funding for NNRs — only 300 million CNY for basic construction and 300 million CNY for capacity-building annually, which can only meet one-third of the needs of the reserves, resulting in a severe lack of conservation funds.

Jianbo Sun summarized the management dilemma of nature reserves in China:

(1) Conservation funds are limited. **The central government only allocates 600 million CNY annually for the existing national nature reserves, averaging only 1.26 million CNY per reserve.** Some reserves do not have the funds to build facilities such as boundary markers and patrol stations.

(2) Inadequate funding renders some reserve institutions functionally impaired, while personnel shortages critically undermine conservation policy implementation.

(3) Due to insufficient funding, conservation methods are outdated, and there is a severe lack of technical support.

(4) Compelled by funding shortages, many regions outsource nature reserve management to commercial tourism operators, transforming them into tourism destinations that directly contradict their conservation objectives.

(5) International exchange and compliance capabilities are “constrained.”

Then, how much money is needed to effectively protect these nature reserves and safeguard China’s ecological security bottom line each year?

The research team led by Yan Xie, deputy researcher at the Institute of Zoology of the Chinese Academy of Sciences and the general coordinator of the research group for nature conservation legislation, previously conducted calculations and provided an answer of 26 billion CNY. “Such a trivial investment can effectively protect 17.48% of China’s land and 10% of its marine areas, thus safeguarding the ecological security bottom line of our country,” Yan Xie told the reporter.

3. Financial Statements of NNR administrations: An Example

Several NNR management administrations disclose funds received from higher level of authorities, including the central government, in their annual financial statements. However, many receive minimal or no funding. For instance, in 2022, the Jinhua and Jiuwan Mountain NNR Management Administration received no transfer payments, while the Daming Mountain NNR Management Administration in Guangxi received only 127,700 CNY ([Figure B5](#)).

FIGURE B5: FINANCIAL STATEMENTS OF SOME MANAGEMENT ADMINISTRATIONS OF NATIONAL NATURE RESERVE IN 2022

Panel A Jinhua National Nature Reserve	
项目 Item	金额 Amount (10 thousand)
一、一般公共预算财政拨款收入 General Public Budgetary Financial Appropriation Income	2,154.35
二、政府性基金预算财政拨款收入 Government Fund Budgetary Financial Appropriation Income	0.00
三、国有资本经营预算财政拨款收入 State-Owned Capital Operation Budgetary Financial Appropriation Income	0.00
四、上级补助收入 Subsidy Income from Higher Authorities	0.00
五、事业收入 Operating Income	0.00
六、经营收入 Business Income	0.00
七、附属单位上缴收入 Income Turned Over by Affiliated Departments	0.00
八、其他收入 Other Income	182.89
Panel B Jiuwan Mountain National Nature Reserve	
项目 Item	金额 Amount (10 thousand)
一、一般公共预算财政拨款收入 General Public Budgetary Financial Appropriation Income	2,840.51
二、政府性基金预算财政拨款收入 Government Fund Budgetary Financial Appropriation Income	0.00
三、国有资本经营预算财政拨款收入 State-Owned Capital Operation Budgetary Financial Appropriation Income	0.00
四、上级补助收入 Subsidy Income from Higher Authorities	0.00
五、事业收入 Operating Income	0.00
六、经营收入 Business Income	0.00
七、附属单位上缴收入 Income Turned Over by Affiliated Departments	0.00
八、其他收入 Other Income	10.15
Panel C Daming Mountain National Nature Reserve	
项目 Item	金额 Amount (10 thousand)
一、一般公共预算财政拨款收入 General Public Budgetary Financial Appropriation Income	9605.15
二、政府性基金预算财政拨款收入 Government Fund Budgetary Financial Appropriation Income)	0.00
三、国有资本经营预算财政拨款收入 State-Owned Capital Operation Budgetary Financial Appropriation Income	0.00
四、上级补助收入 Subsidy Income from Higher Authorities	12.77
五、事业收入 Operating Income	0.00
六、经营收入 Business Income	0.00
七、附属单位上缴收入 Income Turned Over by Affiliated Departments	0.00
八、其他收入 Other Income	363.00

B.5 Political Incentives Exerted on Local Officials by GSA

In 2018, the Ministry of Ecology and Environment (MEE) summoned senior officials from eight cities (prefectures/districts) in Anhui, Chongqing, and Yunnan provinces regarding serious illegal development in NNRs under their jurisdiction. At the meeting’s conclusion, MEE official Changgen Liu acknowledged: “Many issues in China’s nature reserves are historical legacies not caused by those currently in charge. Nevertheless, resolving them now is incumbent upon those responsible.” The summoning of 11 officials at the department level in a single instance was not common in the past and was seen as a strong move by the MEE to address the management of nature reserves. Such direct accountability mechanisms exerted significant political pressure on local governments to implement rectifications.

In addition, according to the interview in [Wang et al. \(2023\)](#), the requirements from the central government during GSA had placed notable political incentives on local officials. For example, one Fujian Forest and Grassland Administration (FGA) official described the requirements as follows:

[The GSA team] compared the satellite images to detect which piece of land had changed, sending all these changes as ‘issue spots’ to us [provincial FGA]. Then we need to find out why the land is changed and recover it. The GSA team will later sample-check our remedies. If some nature reserves do not resolve these issue spots within a timeline, the managers of the nature reserves may be punished by criticism or even demotion or dismissal.

In addition, GSA had played a significant role in raising awareness of ecological protection among local cadres. As one Fujian FGA official stated:

The crackdown on illegal cases by the GSA has served as a wake-up call for local cadres. They now clearly understand that the nature reserve authority enforces stringent regulations, and the designated land use must remain unchanged without exception.

It should be noted that changes in political incentives are to improve the enforcement of local authorities on biodiversity conservation in NNRs, not necessarily associated with an overall significant increase in local political risk, as we discussed in [Section 5.3](#).

B.6 Challenges Encountered During the Implementation of GSA

Local officials faced significant resistance while implementing rectification work under GSA requirements. Wang *et al.* (2023) highlight the substantial challenges encountered during enforcement in nature reserves. Many residents had lived in Fujian Province’s designated reserve areas long before their establishment, making the strict “no human presence” policy in core protected zones operationally impractical, as noted by one official’s complaint:

When the GSA team compared satellite images with the original nature reserve plan, many issue spots emerged. As when the nature reserve was established decades ago, mapping technology was backward, and the boundaries were unclear. Villages, roads, farmland, and commercial forests owned by local farmers were not demarcated from reserve boundaries. Now the GSA team said that ‘no human activities are allowed within the reserve.’ How is it possible to remedy that?

In addition, the long-term lack of adequate management resources compounded the issue. For instance, in Fujian Province, there were two NNRs where the management team consisted of fewer than ten people responsible for over 10,000 hectares, underscoring the deficiency in personnel for proper reserve management.

Beyond official complaints, local residents actively resisted rectification efforts. Most maintained unequivocal opposition, summarized by one village leader: “We understand ecological protection, but we need to live.” Conflicts arose when GSA officials visited the “issue spot”. One GSA official mentioned that they were blocked by residents on their way to the village and were questioned by local farmers:

You guys [GSA officials] came here by car, and you [GSA officials] know it is convenient to drive, don’t we [local villagers] know that? We [local villagers] raised money and built a road, and the government did not pay one penny. But now you [GSA officials] told me that the road is illegal? And ask to demolish it? Is that reasonable?

B.7 Government Spending Features in GSA Implementation: A Case from Jinyun Mountain NNR

The ecological restoration of Jinyun Mountain National Nature Reserve (hereafter Jinyun Mountain NNR) offers a valuable case study for analyzing government expenditure patterns during GSA. The rectification process exhibits four key characteristics: First, rectification activities experienced a delayed start, with real actions by local authorities commencing at least one year after the central government launched the GSA. Second, local governments exhibited a strong commitment to fulfilling GSA mandates through the implementation of significant measures. Third, the rectification efforts and associated expenditures unfolded as a progressive, ongoing process rather than a one-time event. Fourth, expenditures were primarily allocated to three core areas: the demolition of illegal constructions, the relocation of indigenous residents, and ecological restoration efforts, with only a small portion managed through procurement processes.

The following outlines the actions undertaken by the local government in the Jinyun Mountain NNR, along with a back-of-envelope breakdown of the associated spending.

1. Timeline: Accountability from Beijing and local government's actions

In **July 2017**, the Ministry of Environmental Protection of PRC, later renamed the Ministry of Ecology and Environment (MEE), announced the launch of GSA.

In **April 2018**, MEE remote sensing detected over 500 commercial activity sites, including tourist facilities and industrial land, within the Jinyun Mountain NNR, in violation of nature reserve management laws.

In **June 2018**, the central government issued a critical directive concerning the Jinyun Mountain NNR, mandating local authorities implement thorough rectification measures.

In **September 2018**, the MEE summoned officials from relevant government departments responsible for encroachment and destruction incidents within the reserve. Attendees included officials from Chongqing Forestry Bureau, Shapingba District and Beibei District (both administrative districts of Chongqing Municipality). Faced with heightened political pressure from Beijing, the Chongqing Municipal Government leadership elevated addressing reserve issues as a priority.⁴²

In **February 2019**, Chongqing issued the “Guiding Opinions on Implementing Ecological Relocation Pilot Projects in the Jinyun Mountain Nature Reserve,” successively launching pilot relocation initiatives for the indigenous people within the reserve’s core

⁴²From **June 2018**, Min-er Chen, Secretary of the Chongqing Municipal CPC Committee, and Liangzhi Tang, Deputy Secretary of the Municipal CPC Committee and Mayor, conducted nine inspections of Jinyun Mountain to oversee rectification efforts. The government convoked 33 coordination meetings, disciplined 78 officials and 4 responsible departments.

and buffer zones.

In **June 2019**, the government launched a comprehensive ecological restoration program for the reserve, focusing on restoring wetland, increasing vegetation cover, and recovering biodiversity, and began demolishing all resort houses.

During 2018-2020, Chongqing had allocated a cumulative total of 2.895 billion CNY. The funds were primarily directed toward three key areas: the demolition of illegal constructions, the relocation of indigenous residents, and ecological restoration efforts. By the end of 2020, the Jinyun Mountain NNR had achieved significant progress, including the demolition of 88,000 square meters of illegal constructions, the resolution of 269 prominent environmental issues, the planting of 774,000 trees, and the ecological restoration of approximately 100,000 square meters of land.⁴³

During 2021-2022, Chongqing continued to advance ecological restoration efforts (e.g., establishing the forest fire prevention system and the routine inspection system).

2. A back-of-envelope breakdown of expenditures for the GSA

As introduced in previous sections, the expenditures were primarily directed toward three key areas: the demolition of illegal constructions, the relocation of indigenous residents, and ecological restoration efforts.

(1) *Relocation of indigenous residents.* The governments of Beibei District, Shapingba District, and Bishan District, which intersect with the NNR, were responsible for estimated direct relocation compensation payments of approximately 450 million CNY, 80 million CNY, and 50 million CNY, respectively.⁴⁴ Additionally, the government invested 220 million CNY in supporting relocated residents with household registration, healthcare, education, and social insurance services at their new residences.⁴⁵ In total, local governments spent at least 800 million CNY on resident relocation, an expenditure category excluded from government procurement activities.

(2) *Demolition of illegal constructions and ecological restoration.* Detailed public expenditure data for the remaining two categories—demolition of illegal constructions and ecological restoration—are less available. However, partial approximate estimates can be derived. For instance, planting 774,000 trees, assuming an average cost of 1,000 CNY per tree, suggests a minimum cost of 700 million CNY for tree planting alone. Additionally, Beibei District reported the creation of over 300 new jobs, including rangers, checkpoint guards, foresters, and cleaners, with monthly salaries of at least 3,000 CNY. Over three years, the wage expenditure for these personnel would amount to at least 10

⁴³See at <https://mp.weixin.qq.com/s/uTUhvOaeaSzBTewSa-KQgA>.

⁴⁴See at <https://mp.weixin.qq.com/s/ujr5mezJ9t6JG8cTk07fAA> and <https://mp.weixin.qq.com/s/H1BH4NwyI2hZpxKHk0mqbA>.

⁴⁵See at <https://mp.weixin.qq.com/s/6ozty5TsLlN7zRuZi9ovIA>.

million CNY. Collectively, expenditures on resident relocation, tree planting, and new job creation account for approximately half of the total 2.895 billion CNY investment. The remaining half likely covers additional ecological restoration activities (e.g., constructions of biodiversity monitoring and assurance services) and the demolition of illegal constructions, both of which entail significant costs. These activities also involve labor costs for government staff, potentially accompanied by salary adjustments or subsidies.

3. The institutional rationale for the relative magnitude of local public procurement in GSA Compliance

We compiled statistics on government procurement related to Jinyun Mountain during this period. These government procurement deals had a total value of approximately 38.8 million CNY, representing 1.3% of the total 2.895 billion CNY investment. This low proportion of government procurement is consistent with institutional practices, as not all government purchases require public disclosure. Only specific products or services (e.g., machinery, software, technical services, or large-scale constructions) or transactions exceeding certain thresholds (e.g., 0.5 or 1 million CNY in some regions) mandate standard bidding processes and disclosure on government procurement websites. Furthermore, the urgency of NNR restoration tasks and the complexity of procurement processes may incentivize local governments to bypass procurement form to meet conservation goals efficiently. Consequently, the identified procurement expenditure represents a lower bound.

The analysis suggests that government procurement constitutes only a small fraction of total biodiversity expenditures in GSA. Activities such as ecological compensation for trees planting, labor hiring, resident relocation subsidies, and scientific surveys are typically exempt from procurement processes. Additionally, small-scale, fragmented transactions or those not on designated procurement lists may be intentionally or unintentionally excluded from procurement due to the administrative burden of oversight, disclosure, and documentation. These facts elucidate the rationale behind the limited proportion of procurement within the total expenditure for local GSA compliance, complementing the evidence from full-sample procurement variations, which consistently demonstrate a systemic increase in biodiversity conservation spending across NNR municipalities, as discussed in [Section 5.1.1](#).